Plant and Animal Biotechnology

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SPECIAL FEATURES AND ORGANIZATION OF PLANT CELLS

Both plants and animals are made of cells. Plants and animals share many similar characteristics and many of the same organelles. An **organelle** is a specialized unit within the cell that performs a specific function. Organelles act in much the same way our heart, liver, and lungs do. Each organ in our body has a specific function that it carries out that helps us survive. Organelles act in the same way. Some of the organelles that both plant and animal cells share include nuclei, ribosomes, and golgi bodies. There are, however, major differences between plant and animal cells in both their structure and their organelles. The three main differences are cell walls, vacuoles, and chloroplasts.

CELL WALL

The cells of all plants, bacteria and fungi have a rigid, protective covering outside the plasma membrane called **cell wall**. The presence of cell wall in plant cells distinguishes them from animal cells. Among the vascular plants only certain cells connected with the reproductive processes, are naked, all other cells have walls. The cell wall was first observed by **Hooke** in the year 1865 in cork cells. Originally it was thought that the cell wall was a non-living secretion of the protoplasm, but now it is known to be metabolically active and is capable of growth and at least during its growth, contains protoplasmic material.

STRUCTURE OF THE CELL WALL

A typical plant cell has the following three parts.

- Middle lamella
- Primary wall
- Secondary Wall

CHEMICAL COMPOSITION

The chemical composition of cell wall varies in different kingdoms. In bacteria the cell wall is composed of **peptidoglycan,** in Fungi it is made up of **chitin.** The plant cell wall is made up of **cellulose.** Besides cellulose certain other chemicals such as hemicellulose, pectin, lignin, cutin, suberin, silica may also be seen deposited on the wall.

MIDDLE LAMELLA

It is a thin amorphous cement like layer between two adjacent cells. Middle lamella is the first layer, which is deposited at the time of cytokinesis. It is optically inactive (isotropic). It is made up of calcium and magnesium pectates. In addition to these substances proteins are also present.

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PRIMARY WALL

It is the first formed wall of the cell which is produced inner to the middle lamella. It is thin, elastic and extensible in growing cells. It is optically active (anisotropic). It grows by addition of more wall material within the existing one. Such a growth is termed as **intussusception.** Some cells like the parenchymatous cells and meristematic cells have only the primary wall. The primary wall consists of a loose network of cellulose **microfibrils** embedded in a gel like matrix or ground substances. In most of the plants the micro fibrils are made up of cellulose. The micro fibrils are oriented variously according to shape and thickness of the wall. The matrix of the primary wall in which the micro fibrils are embedded is mainly composed of water, hemicellulose, pectin and glycoprotein. Pectin is the filling material of the matrix. Hemicellulose binds the microfibrils with the matrix and the glycoproteins control the orientation of the microfibrils.

SECONDARY WALL

A thick secondary wall is laid inner to the primary wall after the cell has reached maturity. It is laid down is succession of at least three layers often named **S1, S2** and **S3.** It grows in thickness by accretion (apposition) i.e deposition of materials over the existing structures. The central layer (**S2**)is usually the thickest layer. In some cells however, the number of layers may be more than three. The formation of secondary wall is not uniform in all the cells. This results in the differentiation of various types of cells, such as parenchyma, collenchyma, fibres and tracheids.

PLASMODESMATA

The cell wall is not totally complete around the cell. It is interrupted by narrow pores carrying fine strands of cytoplasm, which interlink the contents of the cells. They are called **plasmodesmata.** They form a protoplasmic continuum called **symplast.** It consists of a canal, lined by plasma membrane. It has a simple or branched tubule known as **desmotubule.** Desmotubule is an extension of endoplasmic reticulum. Plasmodesmata serves as a passage for many substances to pass through. It is also believed that they have a role in the relay of stimuli.

PITS

Pits are the areas on the cell wall on which the secondary wall is not laid down. The pits of adjacent cells are opposite to each other. Each pit has a **pit chamber** and a **pit membrane.** The pit membrane consists of middle lamella and primary wall. Pit membrane has many minute pores and thus they are permeable.

Pits are of two types:-

1. **Simple pits**
2. **Bordered pits.**

In simple pits the width of the pit chamber is uniform. There is no secondary wall in the simple pit.

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In bordered pit the secondary wall partly overhangs the pit.

Pits help in the translocation of substances between two adjacent cells. Generally each pit has a **complementary pit** lying exactly opposite to it in the wall of the neighbouring cell. Such pits form a morphological and functional unit called the **pit pair.**

FUNCTIONS OF CELL WALL

- It gives definite shape to the cell.
- It protects the internal protoplasm against injury.
- It gives rigidity to the cell
- It prevents the bursting of plant cells due to endosmosis.
- The walls of xylem vessels, tracheids and sieve tubes are specialized for long distance transport.
- In many cases, the cell wall takes part in offense and defense.

CELL MEMBRANE

All the prokaryotic and eukaryotic cells are enclosed by an elastic thin covering called **plasma membrane.** It is selectively permeable since it allows only certain substances to enter or leave the cell through it. In addition to this eukaryotic cells possess intracellular membranes collectively called cytoplasmic membrane system, that surround the vacuole and cell organelles. Plasma membrane and the sub-cellular membranes are together known as **biological membranes.**

ULTRA STRUCTURE OF THE CELL MEMBRANE

Cell membranes are about 75A° thick. Under the electron microscope they appear to consist of 3 layers.

- An outer electron dense layer of about 20A° thick
- An inner electron dense layer of about 20A° thick.
- A middle pale coloured layer about 35A° thick.

The outer and inner layers are formed of protein molecules whereas the middle one is composed of two layers of phospholipid molecules. Such a trilaminar structure is called “Unit

membrane” which is a basic concept of all membranes.

FLUID MOSAIC MODEL

Many models have been proposed to explain the molecular structure of plasma membrane. Fluid mosaic model was proposed by **Singer and Nicholson** (1972) and it is widely accepted by all. According to this model the cell membrane has **quasifluid** structure. All cellular membranes line

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closed compartments and have a **cytosolic** and an **exoplasmic** face. Membranes are formed of lipids and proteins. According to this model the membrane is viewed as a two dimensional mosaic of phospholipids and protein molecules.

LIPIDS

The lipid molecules form a continuous bilayer. The protein molecules are arranged as e**xtrinsic proteins** on the surface of lipid bilayer and as **intrinsic proteins** that penetrate the lipid bilayer either wholloy or partially. The lipid bilayer is formed of a double layer of phospholipid molecules. They are **amphipathic** molecules i.e. they have a hydrophilic and hydrophobic part. The arrangement of phospholipids forms a water resistant barrier. So that only lipid soluble substances can pass through readily but not water soluble substances.

The phospholipid bilayer forms the basic structure of all biomembranes which also contain proteins, glycoproteins, cholesterol and other steroids and glycolipids. The presence of specific sets of membrane proteins permits each type of membrane to carryout distinctive functions.

PROTEINS

Proteins are arranged in two forms.

- **Extrinsic or peripheral proteins:** These are superficially attached to either face of lipid bimolecular membrane and are easily removable by physical methods.
- **Intrinsic or Integral proteins**: These proteins penetrate the lipid either wholly or partially and are tightly held by strong bonds. In order to remove them, the whole membrane has to be disrupted. The integral proteins occur in various forms and perform many functions.

FUNCTIONS OF PLASMA MEMBRANE

In all cells the plasma membrane has several essential functions to perform. These include transporting nutrients into and metabolic wastes out of the cell preventing unwanted materials from entering the cell. In short, the intercellular and intra cellular transport is regulated by plasma membrane. The plasma membrane maintains the proper ionic composition pH(~7.2) and osmotic pressure of the cytosol. To carry out all these functions, the plasma membrane contains specific transport proteins that permit the passage of certain small molecules but not others. Several of these proteins use the energy released by **ATP** hydrolysis to pump ions and other molecules into or out of the cell against concentration gradients. Small charged molecules such as ATP and amino acids can diffuse freely within the cytosol but are restricted in their ability to leave or enter it across the plasma membrane.

In addition to these universal functions, the plasma membrane has other important functions to perform. Enzymes bound to the plasma membrane catalyze reactions that would occur with difficulty in an aqueous environment. The plasma membranes of many types of eukaryotic cells also contain **receptor proteins** that bind specific **signalling molecules** like hormones, growth factors etc. leading to various cellular responses.

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NOTE:

ENDOCYTOSIS AND EXOCYTOSIS

Endocytosis and exocytosis are active processes involving bulk transport of materials through membranes, either into cells (endocytosis) or out of cells (exocytosis).

Endocytosis occurs by an in folding or extension of the plasma membrane to form a vesicle or vacuole or vauole. It is of two types.

- Phagocytosis: (cell eating)-Substances are taken up in solid form. Cells involving in this process are called phagocytes and said to be phagocytic. (eg.) some white blood cells. A phagocytic vacuole is formed during the uptake.
- Pinocytosis (cell drinking)-Substances are taken up in liquid form. Vesicles which are very small are formed during intake. Pinocytosis is often associated with amoeboid protozozns, and in certain kidney cells involved in fluid exchange. It can also occur in plant cells.

Exocytosis is the reverse of endocytosis by which materials are removed from cells such as undigested remains from food vacuoles.

CELL ORGANELLES

The internal architecture of cells and central metabolic pathways are similar in all plants, animals and unicellular eukaryotic organisma (eg. Yeast) All eukaryotic cells contain a membrane bound nucleus and numerous other organelles in their cytosol. Unique proteins in the interior and membranes of each type of organelle largely determine its specific functional characteristics.

A Typical plant cell contains the following organelles and parts:

1. Mitochondria

They are bounded by two membranes with the inner one extensively folded. Enzymes in the inner mitochondrial membrane and central matrix carry out terminal stages of sugar and lipid oxidation coupled with ATP synthesis.

2. Chloroplasts

They are the sites of Photosynthesis. They are found only in plant cells. They are surrounded by an inner and outer membrane, a complex system of **thylakoid** membranes in their interior contains the pigments and enzymes that absorb light and produce ATP.

3. Nucleus

It is surrounded by an inner and outer membrane. These contain numerous pores through which materials pass between the nucleus and cytosol. The outer nuclear membrane is continuous with the rough endoplasmic reticulum. The nuclear membrane resembles the plasma membrane in its

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function. The nucleus mainly contains DNA organized into linear structures called **chromosomes.**

4. Endoplasmic reticulum

These are a network of inter connected membranes. Two types of Endoplasmic Reticulum are recognised.

- Rough E.R
- Smooth E.R

Rough ER

In this kind of ER, ribosomes are present on the surface. The endoplasmic reticulum is responsible for protein synthesis in a cell. Ribosomes are sub organelles in which the amino acids are actually bound together to form proteins. There are spaces within the folds of ER membrane and they are known as **Cisternae.**

Smooth ER

This type of ER does not have ribosomes.

5. Golgi Body or Golgi Apparatus(G.A.) (Dictyosomes)

Golgi body is a series of flattened sacs usually curled at the edges. Proteins which were formed on ribosomes of rough endoplasmic reticulum are processed in G.A. After processing, the final product is discharged from the G.A. At this time the G.A. bulges and breaks away to form vesicle known as **secretory vesicle.** The vesicles move outward to the cell membrane and either insert their protein contents in the membrane or release these contents outside the cell.

6. Vacuoles

The Vacuoles form about 75% of the plant cell. In the vacuole the plant stores nutrients as well as toxic wastes. If pressure increases within the vacuole it can increase the size of the cell. In this case the cell will become swollen. If the pressure increases further the cell will get destroyed.

7. Ribosomes

Ribosomes are found in cells, both prokaryotic and eukaryotic except in mature sperm cells and RBCs. In eukaryotic cells they occur freely in the cytoplasm and also found attached to the outer surface of rough ER. Ribosomes are the **sites of protein synthesis.**

8. Plasma Membrane

In all the cells the plasma membrane has several functions to perform. These include transporting nutrients into and metabolic wastes out of the cell. It is formed of lipids and proteins.

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9. Microbodies

These are spherical organelles bound by a single membrane. They are the sites of glyoxylate cycle in plants.

10. Cell wall

The cells of all plants have cell wall. It has three parts. 1. Middle lamella 2. Primary wall 3. Secondary wall. It gives definite shape to the plant cell.

NUCLEUS

Nucleus is the largest organelle in eukaryotic cells. It is surrounded by two membranes. Each one is a phospholipid bilayer containing many different types of proteins. The inner nuclear membrane defines the nucleus itself. In many cells the outer nuclear membrane is continuous with the rough ER and the space between the inner and outer nuclear membrane is continuous with the lumen of the rough ER. The two nuclear membranes appear to fuse at the nuclear pores. These ring like pores are constructed of a specific set of membrane proteins and these act like channels that regulate the movement of substances between thenucleus and the cytosol.

In a growing of differentiating cell, the nucleus is metabolically active, producing DNA and RNA. The RNA is exported through nuclear pores to the cytoplasm for use in protein synthesis. In 'resting' cells, the nucleus is inactive or dormant and minimal synthesis of DNA and RNA takes place. In a nucleus that is not dividing, the chromosomes are dispersed and not thick enough to be observed in the light microscope. Only during cell division the chromosomes become visible by light microscopy. **Chromosomes** form the physical basis of heredity. **Genes,** the chemical basis of heredity, are arranged in linear fashion on the chromosomes. A sub organelle of the nucleus, the **nucleolus** is easily recognized under light microscope. Most of the ribosomal RNA of a cell is synthesized in the nucleolus. The finished or partly finished ribosomal sub units pass through a nuclear pore into the cytosol.

The non-nucleolar region of the nucleus is called the **nucleoplasm.** It has very high DNA concentration. Fibrous proteins called lamins form a two dimensional network along the inner surface of the inner membrane giving it shape and apparently binding DNA to it. During the early stages of cell division breakdown of this network occurs.

Functions of Nucleus

- It controls all the metabolic activities of the cell by controlling the synthesis of enzymes required.
- Nucleus controls the inheritance of characters from parents to offspring.
- Nucleus controls cell division.

MITOCHONDRIA

A Mitochondrion is also called as the **“Power house of the cell”** because it stores and releases the energy of the cell. The energy released is used to form ATP (Adenosine Triphosphate)

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Mitochondria are the principal sites of ATP production in aerobic cells. Most eukaryotic cells contain many mitochondria, which occupy up to 25 percent of the volume of the cytoplasm. These complex organelles are among the largest organelles generally exceeded in size only by the nucleus, vacuoles and chloroplasts. Typically the mitochondria are sausage-shaped but these may be granular, filamentous, rodshaped, spherical or thread like. Mitcohondria contain two very different membranes an outer one and an inner one, separated by the inter membrane space. The outer membrane is composed of about half lipid and half protein. The inner membrane is less permeable. It is composed of about 20 percent lipid and 80 percent protein. The surface area of the inner membrane is greatly increased by a large number of infoldings, or **cristae** that protrude into the matrix.

Structure of the Cristae Membrane

The inner of the cristae membrane (i.e. the surface towards the matrix) is covered with numerous (infinite) stalked particles. These are called **F1 Particles, elementary particles** or **sub units.** These particles project into the matrix. Each F1 particle has 3 parts, viz, the head piece, the stalk and the base piece. The respiratory chain consists of enzymes and co-enzymes which constitute the **Electron Transport System, (ETS)** in the mitochondrion. These enzymes and co-enzymes of the ETS act as the electron acceptors in the aerobic respiration reaction. (Oxidative Phosphorylation).

Mitochondria As Semi-autonomous Organelles

Mitochondria are self-perpetuating semi-autonomous bodies. These arise new by the division of existing mitochondria. These are also regarded as intra cellular parasitic prokaryotes that have established symbiotic relationship with the cell. The mitochondrial matrix contains DNA molecules which are circular and 70s ribosomes, tRNA and enzymes for functioning of mitochondrial genes.

PLASTIDS

Plastids are the largest cytoplasmic organelles bounded by double membrane. These are found in most of the plant cells and in some photosynthetic protists. These are absent in prokaryotes and in animal cells. Plastids are of three types namely **chloroplasts, chromoplasts** and **leucoplasts.** Chromoplasts are coloured plastids other than green. They are found in coloured parts of plants such as petals of the flower, pericarp of the fruits etc.

Leucoplasts are the colourless plastids. These colourless plastids are involved in the storage of carbothydrates, fats and oils and proteins. The plastids which store carbohydrates are called amyloplasts. The plastids storing fats and oils are called elaioplasts. The plastids storing protein are called proteinoplasts.

CHLOROPLAST

Chloroplasts can be as long as 10ìm and are typically 0.5 - 2.0 ìm thick, but they vary in size and shape in different cells, especially among the algae. Like mitochondrion, the chloroplast is

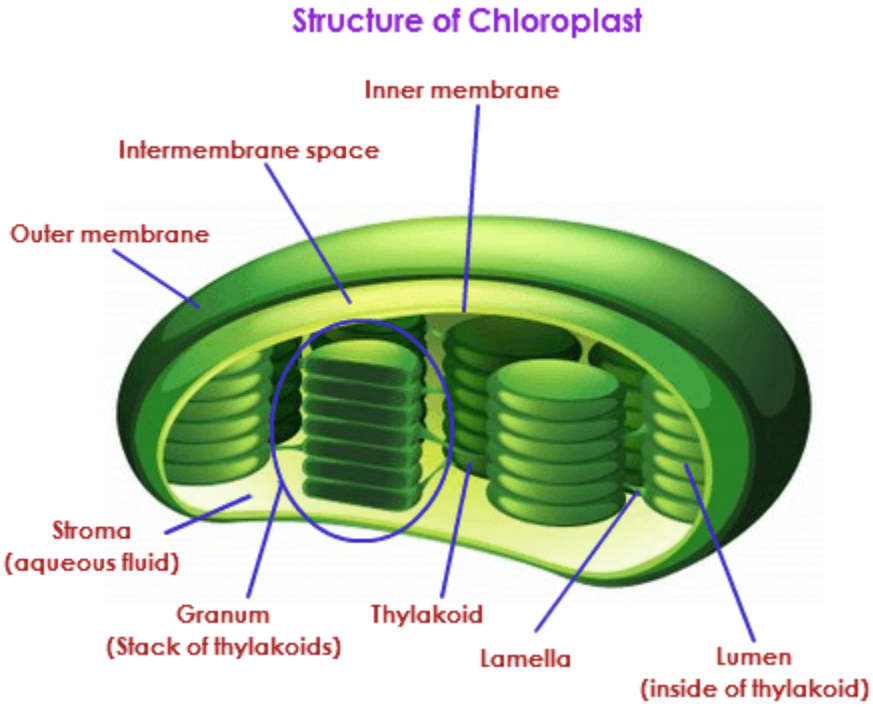
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surrounded by an outer and inner membrane. In addition to this, chloroplasts contain an internal system of extensive inter connected membrane-limited sacs called **thylakoids** which are flattened to form disks. These are often grouped in stakes of 20-50 thylakoids to from what are called **grana** and embedded in a matrix called **stroma**.

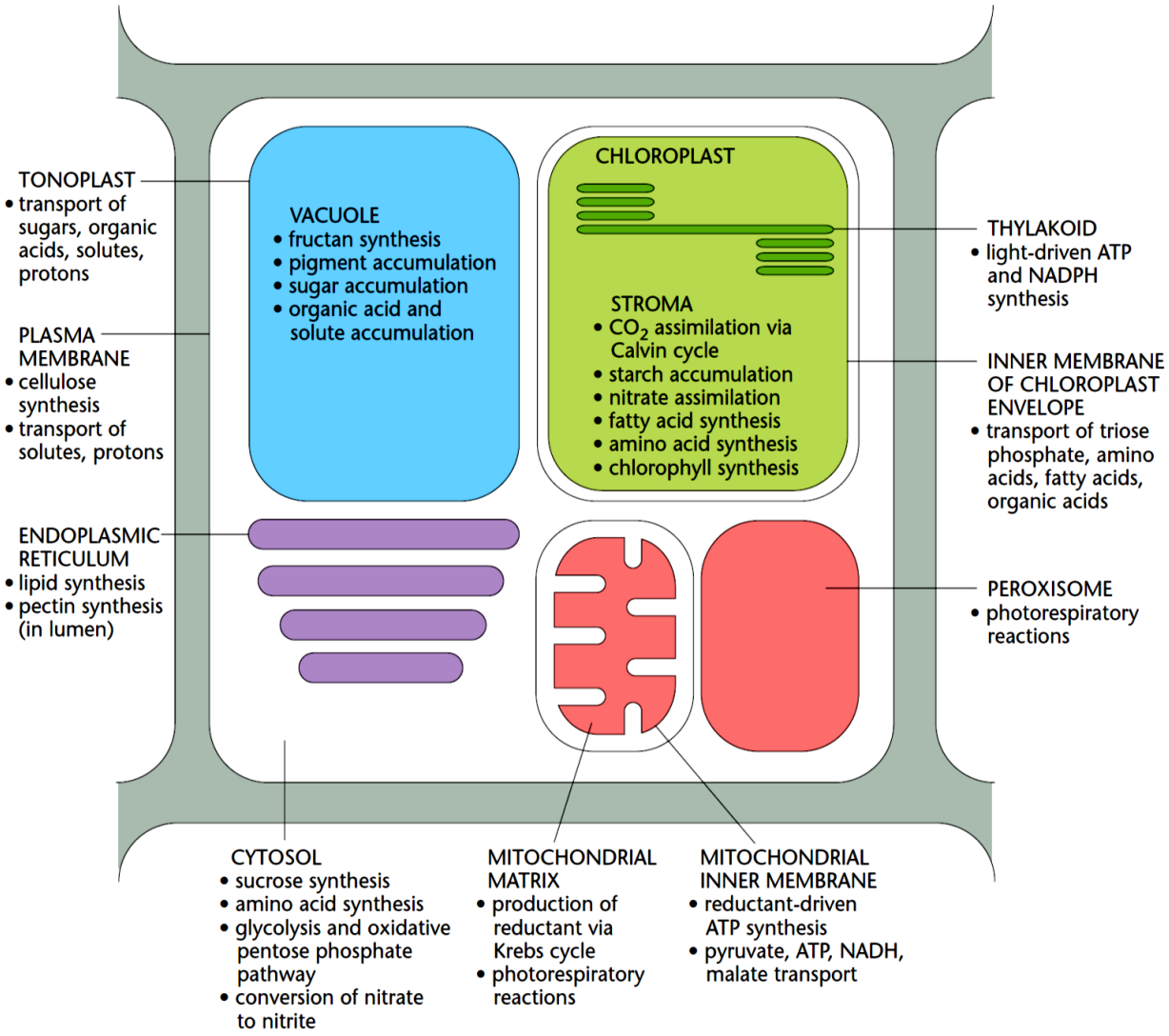
Stroma, a semi fluid, colourless, colloidal complex contains DNA, RNA, ribiosomes and several enzymes. The DNA of chloroplast is circular. The ribosomes are of 70s type. The matrix of higher plant's chloroplasts may contain starch as storage product. Thylakoids may occur attached to the inner membrane of the chloroplast envelop. About 40-100 grana may occur in a chloroplast. Many membranous tubules called stroma lamellae (intergranal thylakoids) interconnect thylakoids of different grana. Thylakoid membrane contains photosynthetic pigments.

The thylakoid membrane contains green pigments (Chlorophylls) and other pigments and enzymes that absorb light and generate ATP during photosynthesis. Part of this ATP is used by enzymes located in stroma to the convert CO2 into three carbon (3C) intermediates which are then exported to the cytosol and converted to sugars.

The molecular mechanism by which ATP is formed is very similar in mitochondria and chloroplasts. Chloroplasts and mitochondria have other features also in common. Both migrate often from place to place within cells and both contain their own DNA which code for some of the key organellar proteins. These proteins are synthesized in the ribosomes within the organelle. However, most of the proteins in each of these organelles are encoded in the nuclear DNA and are synthesized in the cytosol. These proteins are then incorporated into the organelles.



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CELL DIVISION IN PLANT CELLS

CELL CYCLE

As we have discussed in the earlier chapter, the cell cycle amazingly follows a regular timing mechanism. Most eukaryotic cells live according to an internal clock, that is, they proceed through a sequence of phases, called the cell cycle.

During the cell cycle DNA is duplicated during the **synthesis (S)** phase and the copies are distributed to the daughter cells during **mitotic (M)** phase. Most growing plant and animal cells take 10-20 hours to double in number and some duplicate at a much slower rate.

A multi cellular organism usually starts its life as a single cell (zygote). The multiplication of this single cell and its descendants determine the growth and development of the organism and this is

achieved by cell division. Cell division is a complex process by which cellular material is equally divided between daughter cells.

Cell divisions in living things are of three kinds. They are

- Amitosis
- Mitosis
- Meiosis

AMITOSIS

It is a simple type of division where the cell contents including nucleus divide into two equal halves by an inwardly growing constriction in the middle of the cell. This type of cell division is common in prokaryotes.

MITOTIC CELL CYCLE

It is represented by DNA duplication followed by nuclear division (Karyokinesis) which in turn is followed by cytokinesis. Mitotic cell division was first described by **W. Flemming** in 1882. In the same year, mitosis in plants was described by **Strasburger.**

Mitotic cell division is said to be diffused, distributed all over the body. Mitotic cell cycle consists of long **interphase** (which is sub divided into **G1, S and G2** phases), a short **M stage** (or mitotic stage, subdivided into prophase metaphase, anaphase and telophase) and **cytokinesis**. The duration of Interphase and M-phase varies in different cells.

Interphase

It is the stage in between two successive cell divisions during which the cell prepares itself for the process by synthesizing new nucleic acids and proteins. Chromosomes appear as chromatin network. Interphase consists of the following three sub stages.

G1 or Gap-1 phase

This phase starts immediately after cell division. The cell grows in size and there is synthesis of new proteins and RNA needed for various metabolic activities of the cell. A non-dividing cell does not proceed beyond G1 phase. The differentiating cells are said to be in **G0** stage.

S-or Synthetic Phase

During this phase there is duplication of DNA. Thus each chromosome now is composed of two sister chromatids.

G2 or Gap-2Phase

The proteins responsible for the formation of spindle fibres are synthesised during this stage.

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MITOSIS

Mitosis is divided into the following 4 sub stages.

- Prophase
- Metaphase
- Anaphase
- Telophase

1. Prophase

The chromatin network begins to coil and each chromosome becomes distinct s long thread like structure. Each chromosome at this stage has two chromatids that lie side by side and held together by centromere. The nucleus gradually disappears. The nuclear membrane also starts disappearing.

2. Metaphase

The disappearance of nuclear membrane and nucleolus marks the beginning of metaphase. The chromosomes become shorter by further coiling. Finally, the chromosomes become distinct and visible under compound microscope. The chromosomes orient themselves in the equator of the cell in such a way that all the **centromeres** are arranged in the equator forming metaphase plate or equatorial plate. Out of the two chromatids of each chromosome, one faces one pole and the other one faces the opposite pole. At the same time spindle fibres arising from the opposite poles are seen attached to the centromeres. The fibres are made up of proteins rich in sulphur containing amino acids. At late metaphase, the **centromeres divide** and now the chromatids of each chromosome are ready to be separated.

3. Anaphase

Division of centromere marks the beginning of anaphase. The spindle fibres start contracting and this contraction pulls the two groups of chromosomes towards the opposite poles. As the chromosomes move toward opposite poles they assume **V or J** or **I** shaped configuration with the centromere proceeding towards the poles with chromosome arms trailing behind. Such variable shapes of the chromosomes are due to the variable position of centromere.

4. Telophase

At the end of anaphase, chromosomes reach the opposite poles and they uncoil, elongate and become thin and invisible. The nuclear membrane and the nucleouls reappear. thus, two daughter nuclei are formed, one at each pole.

CYTOKINESIS

The division of the cytoplasm is called cytokinesis and it follows the nuclear division by the formation of cell wall between the two daughter nuclei. The formation of cell wall begins as a cell plate also known as **phragmoplast** formed by the aggregation of vesicles produced by Golgi

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bodies. These vesicles which contain cell wall materials fuse with one another to form cell membranes and cell walls. Thus, at the end of mitosis, **two identical** daughter cells are formed.

SIGNIFICANCE OF MITOSIS

- As a result of mitosis two daughter cells which are identical to each other and identical to the mother cell are formed.
- Mitotic cell division ensures that the daughter cells possess a genetical identity, both quantitatively and qualitatively.
- Mitosis forms the basis of continuation of organisms.
- Asexual reproduction of lower plants is possible only by mitosis.
- Vegetative reproduction in higher plants by grafting, tissue culture method are also a consequence of mitosis.
- Mitosis is the common method of multiplication of cells that helps in the growth and development of multi-cellular organism.
- Mitosis helps in the regeneration of lost of damaged tissue and in wound healing.
- The chromosomal number is maintained constant by mitosis for each species.

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MEIOSIS

Meiosis is a process of cell division of the reproductive cells of both plants and animals in which the diploid number of chromosomes is reduced to haploid. Meiosis is also known as **reduction division (RD)** since the number of chromosomes is reduced to half. It takes place only in the reproductive cells during the formation of gametes. Meiosis consists of two complete divisions. As a result of this a diploid cell produces four haploid cells.

The two divisions of meiosis are **meiosis I** or heterotypic division and **meiosis II** or homotypic division. The first division is **meiotic** or reductional in which the number of chromosomes is reduced to half and the **second division** is **mitotic** or **equational**.

In all the sexually reproducing organism the chromosome number remains constant generation after generation. During sexual reproduction the two gametes male and female, each having single set of chromosomes (n) fuse to form a zygote. The zygote thus contains twice as many chromosome as a gamete (**n+n=2n**). In these two sets of chromosomes one set is derived from the male parent and the other set from the female parent. This is how diploids come to possess tow identical sets of chromosomes called **homologous chromosomes.**

Meiosis may take place in the life cycle of a plant during any one of the following events.

- At the time of spore formation ie. During the formation of pollen grains in anther and megaspores in ovules.
- At the time of gamete formation
- At the time of zygote germination.

Each meiotic division cycle is divided into same four stages as in mitosis. **Prophase, Metaphase, Anaphase and Telophase.** The name of each stage is followed by **I** or **II** depending on which division of cycle is involved.

Meiosis I

It consists of four stages namely.

- Prophase I
- Metaphase I
- Anaphase I
- Telophase I

Prophase I

It is the first stage of first meiosis. This is the longest phase of the meiotic division. It includes 5 sub stages namely

- Leptotene
- Zygotene

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- Pachytene
- Diplotene
- Diakinesis

1. Leptotene

The word leptotene means **'thin thread'**. The chromosomes uncoil and become large and thinner. Each chromosome consists of two chromatids.

2. Zygotene

Homologous chromosomes come together and lie side by side throughout their length. This is called **pairing** or **synapsis.** The paired chromosomes are now called **bivalents.** The adjacent non-sister chromatids are joined together at certain posints called **chiasmata.**

3. Pachytene

The chromosomes condense further and become very shorter and thicker. They are very distinct now. The two sister chromatids of each homologous chromosome become clearly visible. The bivalent thus becomes a **tetrad** with four chromatids. In the region of chiasmata, segments of non-sister chromatids of the homologous chromosomes are exchanged and this process is called **crossing over.**

4. Diplotene

The homologous chromosomes condense further. They begin to separate from each other except at the chiasmata. Due to this separation the dual nature of a bivalent becomes apparent and hence the name **diplotene.**

5. Diakinesis

The Chromosomes continue to contract. The separation of chromosome becomes complete due to **terminalisation**. The separation starts from the centromeres and goes towards the end and hence the name terminalisation: The nucleolus and nuclear membrane disappear and spindle formation starts.

Metaphase I

The spindle fibres become prominent. The bivalents align on the equatorial plane. Spindle fibres from opposite poles get attached to the centromeres of homologous chromosomes.

Anaphase I

The two chromosomes of each bivalent (with chromatids still attached to the centromere) separate from each other and move to the opposite poles of the cell. Thus, only one chromosome of each homologous pair reaches each pole. Consequently at each pole only half the number of chromosomes (haploid) is received. These chromosomes are, however not the same as existed at

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the beginning of prophase. Each chromosome consists of one of its original chromatids and the other has a mixture of segments of its own and a segment of chromatid from its homologue (due to crossing over).

Telephase I

This is the last stage of meiosis I. Reorganization of the chromosomes at poles occurs to form two haploid nuclei. Nuclear membrane and nucleolus re-appear. The spindle disappears. There is no cytokinesis after meiosis I. The second meiotic division may follow immediately or after a short inter phase. The DNA of the two haploid nuclei does not replicate.

Meiosis II

The second meiotic division is very much similar to mitosis.

Prophase II

The events of prophase II are similar to mitotic prophase. Nucleolus and nuclear membrane disappear. Spindle fibres are formed at each pole.

Metaphase II

Chromosomes move to the centre of the equatorial plane. They get attached to spindle fibres centromere.

Anaphase II

The sister chromatids separate from one another and are pulled to opposite poles of the spindle due to contraction of the spindle fibres.

Telophase II

The chromosomes begin to uncoil and become thin. They reorganize into nucleus with the reappearance of nucleolus and nuclear membrane in each pole. Cytokinesis follows and **four haploid daughter cells** are formed and thus the meiotic division is completed.

SIGNIFICANCE OF MEIOSIS

- Meiosis helps to maintain the **chromosome number constant** in each plant and animal species. In meiosis four haploid daughter cells are formed from a single diploid cell. This is very important in sexual reproduction during the formation of gametes.
- The occurrence of crossing over results in the **recombination of genes.**
- The recombination of genes results in **genetic variation.**
- The genetic variations form raw materials for **evolution.**

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TOTIPOTENCY

Many somatic plant cells, including some fully differentiated types (e.g. leaf mesophyll), provided they contain intact nuclear, plastid and mitochondrial genomes, have the capacity to regenerate into whole plants. This phenomenon is totipotency, an amazing developmental plasticity that sets plant cells apart from most of their animal counterparts, and was first demonstrated by Steward and Reinert in the 1950s. Often totipotency is revealed when cells or tissues are disturbed or removed from their normal environment and, for example, placed onto artificial media in tissue culture. A differentiated plant cell that is selectively expressing its genetic information can instead initiate expression of the program required for generation of an entire new plant. Many plants have been regenerated from single cells, but not all plant cells are totipotent; some are terminally differentiated, often because of partial or complete genome loss. We can generalise by saying that most plants at most stages of the life cycle have some populations of cells that are totipotent. Totipotency is of course also a property of normal undifferentiated cells, for example in meristems.

The first step in expression of regenerative totipotency is for mature cells to re-enter the cell cycle and resume cell division — a process known as dedifferentiation. This may lead directly to organised development, such as occurs in the epidermal cells of immature hypocotyls of Trifolium (Maheswaran and Williams 1985) where somatic embryos develop (direct embryogenesis), or formation of shoots or roots (direct organogenesis). Alternatively, there may be an intervening callus stage from which organised structures can later be induced to develop — referred to as indirect organogenesis.

Expression of totipotency depends on competence, by which we mean the ability of cells to be induced along a particular developmental pathway, and determination, in which cells become irreversibly committed to a particular pathway. Convolvulus explants display an initial competence to follow two possible developmental pathways — root or shoot formation. Later, once induction of, say, shoots has begun, cells become determined and transfer to conditions that normally induce root formation are now ineffective. However, formation of callus does not necessarily guarantee subsequent organogenesis or that the direction of organogenesis can be controlled. Commitment of root primordia in cereal callus cultures often seems to be irreversible and a high proportion of cells become terminally differentiated root cap cells that secrete the mucus normally associated with caps of intact roots. Calluses which do not lead to regeneration are a common occurrence.

THE CULTURE ENVIRONMENT

When cultured in vitro, all the needs of the plant cells, both chemical (see Table) and physical, have to met by the culture vessel, the growth medium, and the external environment (light, temperature, etc.). The growth medium has to supply all the essential mineral ions required for growth and development. In many cases (as the biosynthetic capability of cells cultured in vitro may not replicate that of the parent plant), it must also supply additional organic supplements

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such as amino acids and vitamins. Many plant cell cultures, as they are not photosynthetic, also require the addition of a fixed carbon source in the form of a sugar (most often sucrose). One other vital component that must also be supplied is water, the principal biological solvent. Physical factors, such as temperature, pH, the gaseous environment, light (quality and duration), and osmotic pressure, also have to be maintained within acceptable limits.

Table: Some elements important for plant nutrition and their physiological function, supplied by the culture medium to support the growth of healthy cultures in vitro:

Element	Function
Nitrogen	Component of proteins, nucleic acids, and some coenzymes; element required in the greatest amounts
Potassium	Regulates osmotic potential; principal inorganic cation
Calcium	Cell-wall synthesis, membrane function, cell signalling
Magnesium	Enzyme cofactor, component of chlorophyll
Phosphorus	Component of nucleic acids; energy transfer; component of intermediates in respiration and photosynthesis
Sulphur	Component of some amino acids (methionine, cysteine) and some cofactors
Chlorine	Required for photosynthesis
Iron	Electron transfer as a component of cytochromes
Manganese	Enzyme cofactor
Cobalt	Component of some vitamins
Copper	Enzyme cofactor; electron-transfer reactions
Zinc	Enzyme cofactor; chlorophyll biosynthesis
Molybdenum	Enzyme cofactor; component of nitrate reductase

PLANT CELL CULTURE MEDIA

Culture media used for the cultivation of plant cells in vitro are composed of three basic components:

- Essential elements, or mineral ions, supplied as a complex mixture of salts;
- An organic supplement supplying vitamins and/or amino acids; and
- A source of fixed carbon; usually supplied as the sugar sucrose.

For practical purposes, the essential elements are further divided into the following categories:

- Macroelements (or macronutrients);

- Microelements (or micronutrients); and
- An iron source.

Complete plant cell culture medium is usually made by combining several different components.

MEDIA COMPONENTS

It is useful to briefly consider some of the individual components of the stock solutions.

MACROELEMENTS

As is implied by the name, the stock solution supplies macroelements required in large amounts for plant growth and development. Nitrogen, phosphorus, potassium, magnesium, calcium, and sulphur (and carbon, which is added separately) are usually regarded as macroelements. These elements usually comprise at least 0.1% of the dry weight of plants. Nitrogen is most commonly supplied as a mixture of nitrate ions (from KNO3) and ammonium ions (from NH4NO3). Theoretically, there is an advantage in supplying nitrogen in the form of ammonium ions, as nitrogen must be in the reduced form to be incorporated into macromolecules. Nitrate ions therefore need to be reduced before incorporation. However, at high concentrations, ammonium ions can be toxic to plant cell cultures and uptake of ammonium ions from the medium causes acidification of the medium. For ammonium ions to be used as the sole nitrogen source, the medium needs to be buffered. High concentrations of ammonium ions can also cause culture problems by increasing the frequency of vitrification (the culture appears pale and 'glassy' and is usually unsuitable for further culture). Using a mixture of nitrate and ammonium ions has the advantage of weakly buffering the medium as the uptake of nitrate ions causes OH^- ions to be excreted. Phosphorus is usually supplied as the phosphate ion of ammonium, sodium, or potassium salts. High concentrations of phosphate can lead to the precipitation of medium elements as insoluble phosphates.

MICROELEMENTS

Microelements are required in trace amounts for plant growth and development, and have many and diverse roles. Manganese, iodine, copper, cobalt, boron, molybdenum, iron, and zinc usually comprise the microelements, although other elements such as nickel and aluminum are found frequently in some formulations.

Iron is usually added as iron sulphate, although iron citrate can also be used. Ethylenediaminetetra-acetic acid (EDTA) is usually used in conjunction with iron sulphate. The EDTA complexes with the iron to allow the slow and continuous release of iron into the medium. Uncomplexed iron can precipitate out of the medium as ferric oxide.

ORGANIC SUPPLEMENTS

Only two vitamins, thiamine (vitamin B1) and myoinositol (considered a B vitamin), are considered essential for the culture of plant cells in vitro. However, other vitamins are often added to plant cell culture media for historical reasons. Amino acids are also commonly included

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in the organic supplement. The most frequently used is glycine (arginine, asparagine, aspartic acid, alanine, glutamic acid, glutamine, and proline are also used), but in many cases its inclusion is not essential. Amino acids provide a source of reduced nitrogen and, like ammonium ions, uptake causes acidification of the medium. Casein hydrolysate can be used as a relatively cheap source of a mix of amino acids.

CARBON SOURCE

Sucrose is cheap, easily available, readily assimilated, and relatively stable, and is therefore the most commonly used carbon source. Other carbohydrates (such as glucose, maltose, galactose, and sorbitol) can also be used and in specialized circumstances may prove superior to sucrose.

GELLING AGENTS

Media for plant cell culture in vitro can be used in either liquid or 'solid' forms, depending on the type of culture being grown. For any culture types that require the plant cells or tissues to be grown on the surface of the medium, it must be solidified (more correctly termed **gelled**).

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REGENERATION OF PLANTS

In broad terms, two methods of plant regeneration are widely used in plant transformation studies: somatic embryogenesis and organogenesis.

SOMATIC EMBRYOGENESIS

In somatic (asexual) embryogenesis, embryo-like structures, which can develop into whole plants in a way analogous to zygotic embryos, are formed from somatic tissues. These somatic embryos can be produced either directly or indirectly. In direct somatic embryogenesis, the embryo is formed directly from a cell or small group of cells without the production of an intervening callus. Though common from some tissues (usually reproductive tissues such as the nucellus, styles, or pollen), direct somatic embryogenesis is generally rare in comparison with indirect somatic embryogenesis.

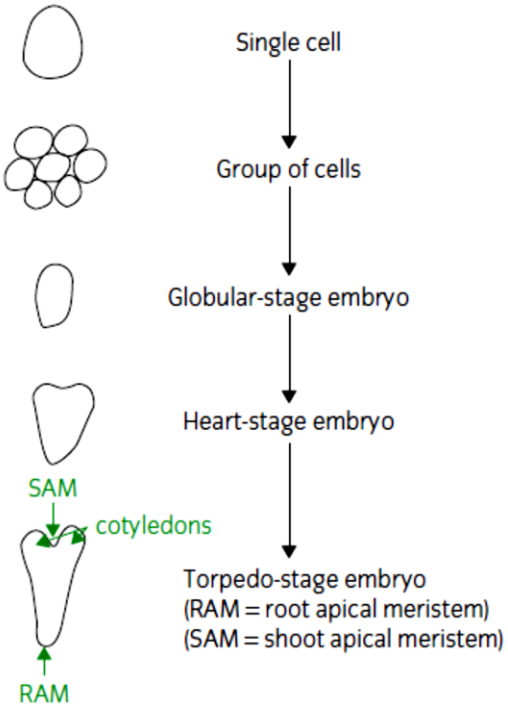


Figure: A schematic representation of the sequential stages of somatic embryo development.

Somatic embryos may develop from single cells or from a small group of cells. Repeated cell divisions lead to the production of a group of cells that develop into an organized structure known as a globular-stage embryo. Further development results in heart- and torpedo-stage embryos, from which plants can be regenerated. Zygotic embryos undergo a fundamentally similar development through the globular (which is formed after the 16-cell stage), heart, and torpedo stages. Polarity is established early in embryo development. Signs of tissue differentiation become apparent at the globular stage and apical meristems are apparent in heart-stage embryos.

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In indirect somatic embryogenesis, callus is first produced from the explant. Embryos can then be produced from the callus tissue or from a cell suspension produced from that callus. Somatic embryogenesis from carrot is the classical example of indirect somatic embryogenesis.

Somatic embryogenesis usually proceeds in two distinct stages. In the initial stage (embryo initiation), a high concentration of 2, 4-D is used. In the second stage (embryo production), embryos are produced in a medium with no or very low levels of 2, 4-D. In many systems, it has been found that somatic embryogenesis is improved by supplying a source of reduced nitrogen, such as specific amino acids or casein hydrolysate.

INDIRECT SOMATIC EMBRYOGENESIS IN CARROT (DAUCUS CAROTA)

A callus can be established from explants from a wide range of carrot tissues by placing the explant on solid medium (e.g. Murashige and Skoog (MS) medium) containing 2, 4-D (1 mg/l). This callus can be used to produce a cell suspension by placing it in agitated liquid MS medium also containing 2, 4-D (1 mg/l). The cell suspension can be maintained by repeated subculturing into 2, 4-D-containing medium. Removal of the old 2, 4-D-containing medium and replacement with fresh medium containing ABA (0.025 mg/l) results in the production of embryos.

DIRECT SOMATIC EMBRYOGENESIS FROM ALFALFA (MEDICAGO FALCATA)

Young trifoliate leaves are used as the explant (see figure). These are removed from the plant and chopped into small pieces. The pieces are washed in a plant growth regulator-free medium and placed in liquid medium (B5) supplemented with 2,4-D (4 mg/l), kinetin (0.2 mg/l), adenine (1 mg/l), and glutathione (10 mg/l). The cultures are maintained in agitated liquid medium for about 10–15 days. Washing the explants and replacing the old medium with B5 medium supplemented with maltose and polyethylene glycol results in the development of the somatic embryos. These somatic embryos can be matured on solid medium containing ABA. Note that in both cases, although the production of somatic embryos from alfalfa necessitates the use of more complicated media, the production of embryos is fundamentally a two-step process. The initial medium, which contains 2,4-D, is replaced with a medium that does not contain 2,4-D.

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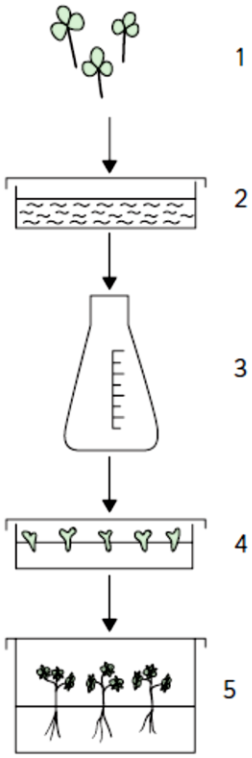


FIG: Direct somatic embryogenesis in alfalfa. (1) Explants are removed from plants grown in vitro. (2) Explants are placed in liquid medium for embryo induction. (3) Embryos develop to the globular stage in liquid medium supplemented with maltose and polyethylene glycol. (4) Embryos mature on gelled medium containing ABA. (5) Embryos develop into plants on solid medium.

ORGANOGENESIS

Somatic embryogenesis relies on plant regeneration through a process analogous to zygotic embryo germination. Organogenesis relies on the production of organs, either directly from an explant or from a callus culture. There are three methods of plant regeneration via organogenesis.

The first two methods depend on adventitious organs arising either from a callus culture or directly from an explant. Alternatively, the third method is by axillary bud formation and growth, which can also be used to regenerate whole plants from some types of tissue culture.

Organogenesis relies on the inherent plasticity of plant tissues, and is regulated by altering the components of the medium. In particular, it is the auxin to cytokinin ratio of the medium that determines which developmental pathway the regenerating tissue will take. It is usual to induce shoot formation by increasing the cytokinin to auxin ratio of the culture medium. These shoots can then be rooted relatively simply.

PRODUCTION OF SOMATIC HYBRIDS AND CYBRIDS

The Somatic cell hybridization/ para sexual hybridization or Protoplast fusion offers an alternative method for obtaining distant hybrids with desirable traits significantly between

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species or genera, which cannot be made to cross by conventional method of sexual hybridization.

SOMATIC HYBRIDIZATION

Somatic hybridization broadly involves in vitro fusion of isolated protoplasts to form a hybrid cell and its subsequent development to form a hybrid plant. The process involves: a) fusion of protoplasts, (b) Selection of hybrid cells, (c) identification of hybrid plants. During the last two decades, a variety of treatments have been used to bring about the fusion of plant protoplasts. Protoplast fusion can be achieved by spontaneous, mechanical, or induced fusion methods. These treatments include the use of fusogens like NaNO3, high pH with high Ca2++ ion concentration, use of polyethylene glycol (PEG), and electro fusion. These inducing agents used in protoplast fusion are called 'fusogen'. PEG treatment is the most widely used method for protoplast fusion as it has certain advantages over others. These are: (a) it results in a reproducible high-frequency of heterokaryon formation, (b) The PEG fusion is non-specific and therefore can be used for a wide range of plants., (c) It has low toxicity to the cell and (d) The formation of binucleate heterokaryons is low.

MECHANISM OF FUSION

The fusion of protoplasts takes place in three phases- agglutination, plasma membrane fusion and formation of heterokaryons. When the two protoplasts come in close contact with each other, they adhere to each other. This agglutination can be induced by PEG, high pH and high Ca2+. The protoplast membranes get fused at localized sites at the point of adhesion. This leads to the formation of cytoplasmic bridges between protoplasts. High pH and high Ca2+ ions neutralize the surface charges on the protoplasts which allow closer contact and membrane fusion between agglutinated protoplasts. The fused protoplasts become round as a result of cytoplasmic bridges which leads to the formation of spherical homokaryon or heterokaryon.

SELECTION OF HYBRID CELLS

The methods used for the selection of hybrid cells are biochemical, visual and cytometric methods using fluorescent dyes. The biochemical methods for selection of hybrid cells are based on the use of biochemical compounds in the medium. The drug sensitivity method is useful for the selection hybrids of two plants species, if one of them is sensitive to a drug. Another method, auxotrophic mutant selection method involves the auxotrophs which are mutants that cannot grow on a minimal medium. Therefore specific compounds are added in the medium. The selection of auxotropic mutants is possible only if the hybrid cells can grow on a minimal medium. The visual method involves the identification of heterokaryons under the light microscope. In some of the somatic hybridizations, the chloroplast deficient protoplast of one plant species is fused with the green protoplast of another plant species. The heterokaryons obtained are bigger and green in colour while the parental protoplasts are either small or colourless. The cytometric method uses flow cytometry and flourescent-activated cell sorting techniques for the analysis of plant protoplasts.

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APPLICATIONS OF SOMATIC HYBRIDIZATION

- **Creation of hybrids with disease resistance** - Many disease resistance genes (e.g. tobacco mosaic virus, potato virus X, club rot disease) could be successfully transferred from one species to another, e.g. resistance has been introduced in tomato against diseases such as TMV, spotted wilt virus and insect pests.
- **Environmental tolerance** - using somatic hybridization the genes conferring tolerance for cold, frost and salt were introduced in e.g. in tomato.
- **Cytoplasmic male sterility** - using cybridization method, it was possible to transfer cytoplasmic male sterility.
- **Quality characters** - somatic hybrids with selective characteristics have been developed, e.g. the production of high nicotine content.

CHROMOSOME NUMBER IN SOMATIC HYBRIDS

The chromosome number in the somatic hybrids is generally more than the total number of both of the parental protoplasts. If the chromosome number in the hybrid is the sum of the chromosomes of the two parental protoplasts, the hybrid is said to be symmetric hybrid. Asymmetric hybrids have abnormal or wide variations in the chromosome number than the exact total of two species.

In 1972, Carlson and his associates produced the first inter-specific somatic hybrid between Nicotiana glauca and N. langsdorffii. In 1978, Melchers and his co-workers developed the first inter-genetic somatic hybrids between Solanum tuberosum (potato) and Lycopersicon esculentum (tomato). The hybrids are known as 'Pomatoes or Topatoes'.

LIMITATIONS OF SOMATIC HYBRIDIZATION

- Somatic hybridization does not always produce plants that give fertile and visible seeds.
- There is genetic instability associated with protoplast culture.
- There are limitations in the selection methods of hybrids, as many of them are not efficient.
- Somatic hybridization between two diploids results in the formation of an amphidiploids which is not favorable therefore haploid protoplasts are recommended in somatic hybridization.
- It is not certain that a specific character will get expressed in somatic hybridization.
- Regenerated plants obtained from somatic hybridization are often variable due to somaclonal variations, chromosomal elimination, organelle segregation etc.
- Protoplast fusion between different species/genus is easy, but the production of viable somatic hybrids is not always possible.

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CYBRIDS

The cytoplasmic hybrids where the nucleus is derived from only one parent and the cytoplasm is derived from both the parents are referred to as cybrids. The process of formation of cybrids is called cybridization. During the process of cybridization and heterokaryon formation, the nuclei are stimulated to segregate so that one protoplast contributes to the cytoplasm while the other contributes nucleus alone. The irradiation with gamma rays and X-rays and use of metabolic inhibitors makes the protoplasts inactive and non-dividing. Some of the genetic traits in certain plants are cytoplasmically controlled. This includes certain types of male sterility, resistance to certain antibiotics and herbicides. Therefore cybrids are important for the transfer of cytoplasmic male sterility (CMS), antibiotic and herbicide resistance in agriculturally useful plants. Cybrids of Brassica raphanus that contain nucleus of B. napus, chloroplasts of atrazinc resistant B. capestris and male sterility from Raphanus sativa have been developed.

IN VITRO PLANT GERMPLASM CONSERVATION

Germplasm refers to the sum total of all the genes present in a crop and its related species. The conservation of germ plasm involves the preservation of the genetic diversity of a particular plant or genetic stock for its use at any time in future. It is important to conserve the endangered plants or else some of the valuable genetic traits present in the existing and primitive plants will be lost. A global organization- International Board of Plant Genetic Resources (IBPGR) has been established for germ plasm conservation and provides necessary support for collection, conservation and utilization of plant geneic resources throughout the world. The germ plasm is preserved by the following two ways:

- **In-situ conservation** - The germ plasm is conserved in natural environment by establishing biosphere reserves such as national parks, sanctuaries. This is used in the preservation of land plants in a near natural habitat along with several wild types.
- **Ex-situ conservation** - This method is used for the preservation of germ plasm obtained from cultivated and wild plant materials. The genetic material in the form of seeds or in vitro cultures are preserved and stored as gene banks for long term use.

In vivo gene banks have been made to preserve the genetic resources by conventional methods e.g. seeds, vegetative propagules, etc. **In vitro** gene banks have been made to preserve the genetic resources by non-conventional methods such as cell and tissue culture methods. This will ensure the availability of valuable germ plasm to breeder to develop new and improved varieties.

The methods involved in the in vitro conservation of germ plasm are:

- **Cryopreservation**- In cryopreservation (Greek-krayos-frost), the cells are preserved in the frozen state. The germplasm is stored at a very low temperature using solid carbon dioxide (at -79^0C), using low temperature deep freezers (at -80^0C), using vapor nitrogen (at- 150^0C) and liquid nitrogen (at-196^0C). The cells stay in completely inactive state and thus can be conserved for long periods. Any tissue from a plant can be used for cryopreservation e.g.

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meristems, embryos, endosperms, ovules, seeds, cultured plant cells, protoplasts, calluses. Certain compounds like- DMSO (dimethyl sulfoxide), glycerol, ethylene, propylene, sucrose, mannose, glucose, praline, ace amide etc are added during the cryopreservation. These are called cryoprotectants and prevent the damage caused to cells (by freezing or thawing) by reducing the freezing point and super cooling point of water.

- **Cold Storage**- Cold storage is a slow growth germ plasm conservation method and conserves the germ plasm at a low and non-freezing temperature (1-9^{0}C). The growth of the plant material is slowed down in cold storage in contrast to complete stoppage in cryopreservation and thus prevents cryogenic injuries. Long term cold storage is simple, cost effective and yields germ plasm with good survival rate. Virus free strawberry plants could be preserved at 10^{0}C for about 6 years. Several grape plants have been stored for over 15 years by using a cold storage at temperature around 9^{0}C and transferring them in the fresh medium every year.

- **Low pressure and low oxygen storage**- In low- pressure storage, the atmospheric pressure surrounding the plant material is reduced and in the low oxygen storage, the oxygen concentration is reduced. The lowered partial pressure reduces the in vitro growth of plants. In the low-oxygen storage, the oxygen concentration is reduced and the partial pressure of oxygen below 50 mmHg reduces plant tissue growth. Due to the reduced availability of O2, and reduced production of CO2, the photosynthetic activity is reduced which inhibits the plant tissue growth and dimension. This method has also helped in increasing the shelf life of many fruits, vegetables and flowers.

The germ plasm conservation through the conventional methods has several limitations such as short-lived seeds, seed dormancy, seed-borne diseases, and high inputs of cost and labour. The techniques of cryo-preservation (freezing cells and tissues at -196^{0}C) and using cold storages help us to overcome these problems.

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SOMACLONAL VARIATION

INTRODUCTION

The genetic variations found in the in vitro cultured cells are collectively referred to as somaclonal variation and the plants derived from such cells are called as 'soma clones'. It has been observed that the long-term callus and cell suspension culture and plants regenerated from such cultures are often associated with chromosomal variations. It is this property of cultured cells that finds potential application in the crop improvement and in the production of mutants and variants (e.g. disease resistance in potato).

Larkin and Scowcroft (1981) gave the term 'soma clones' for plant variants obtained from tissue cultures of somatic tissues. Similarly, if the tissue from which the variants have been obtained is having gametophytic origin such as pollen or egg cell, it is known as 'gametoclonal' variation.

They explained that it may be due to:

- Reflection of heterogeneity between the cells and explant tissue,
- A simple representation of spontaneous mutation rate, and
- Activation by culture environment of transposition of genetic materials.

Somaclonal Variations has been used in plant breeding programmes where the genetic variations with desired or improved characters are introduced into the plants and new varieties are created that can exhibit disease resistance, improved quality and yield in plants like cereals, legumes, oil seeds tuber crops etc. Somaclonal variation is applicable for seed

APPLICATIONS OF SOMACLONAL VARIATIONS

- Methodology of introducing somaclonal variations is simpler and easier as compared to recombinant DNA technology.
- Development and production of plants with disease resistance e.g. rice, wheat, apple, tomato etc.
- Develop biochemical mutants with abiotic stress resistance e.g. aluminium tolerance in carrot, salt tolerance in tobacco and maize.
- Development of somaclonal variants with herbicide resistance e.g. tobacco resistant to sulfonylurea.
- Development of seeds with improved quality e.g. a new variety of Lathyrus sativa seeds (Lathyrus Bio L 212) with low content of neurotoxin.
- Bio-13 – A somaclonal variant of Citronella java (with 37% more oil and 39% more citronellon), a medicinal plant has been released as Bio-13 for commercial cultivation by Central Institute for Medicinal and Aromatic Plants (CIMAP), Lucknow, India.

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- Super tomatoes- Heinz Co. and DNA plant Technology Laboratories (USA) developed Super tomatoes with high solid component by screening soma clones which helped in reducing the shipping and processing costs.

ADVANTAGES

The major likely benefit of somaclonal variation in plant is improvement. Somaclonal variation leads to the creation of additional genetic variability. Characteristics for which somaclonal mutants can be enriched during in vitro culture includes resistance to disease pathotoxins, herbicides and tolerance to environmental or chemical stress, as well as for increased production of secondary metabolites. Micropropagation can be carried out throughout the year independent of the seasons.

DISADVANTAGES

A serious disadvantage of somaclonal variation occurs in operations which require clonal uniformity, as in the horticulture and forestry industries where tissue culture is employed for rapid propagation of elite genotypes. Ways of reducing somaclonal variation: Different steps can be used. It is well known that increasing numbers of subculture increases the likelihood of somaclonal variation, so the number of subcultures in micro propagation protocols should be kept to a minimum. Regular reinitiating of clones from new explants might reduce variability over time. Another way of reducing somaclonal variation is to avoid 2,4-D in the culture medium, as this hormone is known to introduce variation. Vitrification [hyperhydracity] may be a problem in some species. In case of forest trees, mature elite trees can be identified and rapidly cloned by this technique.

High production cost has limited the application of this technique to more valuable ornamental crops and some fruit trees.

PRODUCTION OF VIRUS FREE PLANTS

The viral diseases in plants transfer easily and lower the quality and yield of the plants. It is very difficult to treat and cure the virus infected plants therefore the plant breeders are always interested in developing and growing virus free plants.

In some crops like ornamental plants, it has become possible to produce virus free plants through tissue culture at the commercial level. This is done by regenerating plants from cultured tissues derived from a) virus free plants, b) meristems which are generally free of infection - In the elimination of the virus, the size of the meristem used in cultures play a very critical role because most of the viruses exist by establishing a gradient in plant tissues. The regeneration of virus-free plants through cultures is inversely proportional to the size of the meristem used., c) meristems treated with heat shock (34-360C) to inactivate the virus, d) callus, which is usually virus free like meristems.e) chemical treatment of the media- attempts have been made to eradicate the viruses from infected plants by treating the culture medium with chemicals e.g. addition of cytokinins suppressed the multiplication of certain viruses.

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Among the culture techniques, meristem-tip culture is the most reliable method for virus and other pathogen elimination.

Viruses have been eliminated from a number of economically important plant species which has resulted in a significant increase in the yield and production e.g. potato virus X from potato, mosaic virus from cassava etc. These virus free plants are not disease resistant so there is a need to maintain stock plants to multiply virus free plants whenever required.

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PLANT PRODUCTS OF INDUSTRIAL IMPORTANCE

Phytochemicals, as the word implies, are the individual chemicals from which plants are made.

Phytochemicals of Medicinal Value in Plants: In common usage today, many phytochemicals are associated with health benefits. They have a long history, which continues today, as medicines (Rouhi, 2003b). Many, though not all, of these materials are classified as **secondary metabolites.**

LIPIDS AND DERIVATIVES

Lipids are often defined as water-insoluble biomolecules that are soluble in non polar solvents (Bruice, 2004). This is a convenient definition because it encompasses a large area of chemical space, including many types of compounds that are otherwise hard to classify. There are two problems with this definition. First, given a large enough hydrocarbon (**hydrophobic**) component, most organic compounds could fall within this scope. Second, many of the classical lipids (for example, the fatty acids) have significant solubility in water. A more constricted definition of lipids is to simply classify them as fatty acids and their derivatives, and to treat other hydrocarbon-based natural products separately.

Fatty acids are carboxylic acids that contain a long, hydrocarbon chain. The derivatives of fatty acids may be acyglycerol esters, wax esters,or alcohols such as sterols. Additional acid derivatives include phosphates (glycerophospholipids) orcarbohydrates (glycoglycerolipids).

HYDROCARBONS

Comprising a relatively small group of compounds, the least polar organic natural products are the **hydrocarbons**. Hydrocarbons are simply molecules that contain only hydrogen and carbon atoms. The aliphatic hydrocarbons are straight chain hydrocarbons, usually having an odd number of carbon atoms, resulting from the decarboxylation of their fatty acid counterparts (Savage et al., 1996). Devoid of any heteroatoms, these compounds have relatively simple structures. Hydrocarbons, in general, may be either **saturated** or **unsaturated**— the latter contain multiple bonds. Each double bond results in two fewer hydrogen atoms relative to the saturated counterpart (thus, four fewer hydrogen atoms for triple bonds) and is, therefore, in a higher oxidation state.

They may contain straight chains, branched chains, as well as rings. Being purely organic in nature, they are highly insoluble in water, that is, they are "greasy." With rare exceptions, such as highly halogenated compounds, they are less dense than water. Compounds containing aromatic rings generally show increased stability. Highly aromatic compounds may have reduced solubility in common organic solvents, due to stronger intermolecular interactions.

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SATURATED HYDROCARBONS

Saturated hydrocarbons are the simplest and least polar organic natural products. In living plants, saturated hydrocarbons are universally distributed as the waxy coatings (cuticular waxes) on leaves and as **cuticle** waxes on the surfaces of fruits.

Typical examples include:

- **Nonacosane** CH_3 $(CH_2)_{27}CH_3$; and
- **Hentriacontane** $CH_3(CH_2)_{29}CH_3$. Several plants are rich in aliphatic hydrocarbons used in vegetable oils. For example, **olive oil,** derived from the fruits of olive (Olea europea), contains hydrocarbons ranging from $C_{13}$to C_{28}. Branched simple alkanes (again excluding terpenes) rarely occur in significant quantity in plants.

UNSATURATED HYDROCARBONS

The simplest unsaturated hydrocarbon is ethylene, $H_2C=CH_2$, an important plant hormone. Plant hormones such as ethylene are small organic compounds that influence physiological responses at very low concentrations. Produced by the amino acid methionine, ethylene causes trees to lose their leaves (abscission), stems to thicken, and fruit to ripen. In the latter case, adding low concentrations of ethylene to the air can artificially promote fruit ripening, as with apples (Malus spp.) or pineapple (Ananas comosus). Concentrations as low as 0.01 ppm was shown to distort the growth of tomato and marigold plants, causing what is termed epinasty.

Larger unsaturated hydrocarbons are also common as **plant waxes.** Exceptionally high amounts of alkenes were detected in rye (Secalecereale) pollen, rose (Rosaspp.) petals, and sugarcane (Saccharumspp.). As the chain length and degree of unsaturation increase, the hydrocarbons become waxy and then solid at room temperature. Waxes may be either long-chain hydrocarbons or esters of fatty acids.

POLYACETYLENES

The Polyacetylenes are a unique group of naturally occurring hydrocarbon derivatives characterized by one or more acetylenic groups in their structures. The electronic arrangement of the carbon atoms in a triple bond results in a linear shape for this region of the molecule. Typical polyacetylene soften contain a wide variety of additional functional groups. The domestic carrot (Daucus carota), for example, contains four polyacetylenes, the major one being falcarinol which is a mild neurotoxin found only to be present in 2 mg·kg–1(dry weight) of carrot roots. Other plants, such as the water dropwort (Oenanthe crocata), are commonly found near streams in the Northern Hemisphere and contain several toxicpolyacetylenes and should not be consumed (Hansen and Boll, 1986). The water dropwort (Oenanthe crocata) contains the violent toxin, cicutoxin, which can result in convulsions and respiratory paralysis.

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FUNCTIONALIZED HYDROCARBONS

Excluding the lipids and the terpenes, simple functionalized hydrocarbons are less abundant but not uncommon in plants. Here, we consider these in ascending order from halide, to alcohol and sulfur containing hydrocarbons, then to aldehydes and ketones, stopping just before the fatty acids.

HALOGENATED HYDROCARBONS

A halogen is any of the group 7A elements found on the periodic table of elements (flourine, chlorine, bromine, iodine, or astatine). Although virtually unknown among their terrestrial counterparts, the marine environment has long been recognized as a source for natural products that contain both chlorine and bromine. Iodinated natural products are rare but have been known since the 1970s, and fluorinated natural products were also identified. In the latter case, the source of fluorine in structures such as nucleocidin is believed to be derived from fluoroacetyl Co-A. For the other halogens, haloperoxidases, such as vanadium bromoperoxidase, are the primary biogenetic source.

ALCOHOLS

An alcohol can be any of a class of compounds characterized by the presence of a hydroxyl group (–OH group) covalently bonded to a saturated carbon atom. Large varieties of volatile aliphatic alcohols occur in small concentrations in plants and were classically referred to within the group of essential oils. Their role may be related to their often strong odors, attracting them to insect pollinators and animal seed disseminators. All of the straight-chain alcohols from C1 (methanol) to C10 were found in plants in either free or esterified form. Several larger alcohols, such as ceryl alcohol, $CH_3(CH_2)_{25}OH$, are regular constituents of cuticular waxes. Like the terpenes, the aliphatic alcohols, including cis-3-hexen-1-ol (leaf alcohol), have characteristic and sometimes attractive odors and are of interest to the fragrance industry. The list of alcohols in plants, however, goes on and on, and the reader will notice that the hydroxyl group is associated with many different types of plant molecules.

SULFIDES AND GLUCOSINOLATES

Hydrocarbon sulfides have at least one sulfur atom and are found in relatively few plants. Those that contain them, such as skunk cabbage (Symplocarpus foetidus), are readily recognizable by their obnoxious odors. Sulfides, including the simple hydrocarbon sulfides, are common among the Allium species(onions and their relatives), many of which are lachrymators (substances that make the eyes water) and have pungent odors. Cyclic examples, such as thiophenes, are limited primarily to the Asteraceae (aster or sunflower family) and are found in association with the polyacetylenes (Christenson et al., 1990). The glucosinolates are sulfur-containing natural products primarily from the Brassicaceae (mustard family). They consist of a thioglucose and sulfonated oxime, with a specific side chain for each of the over 100 glucosinolates that have been identified.Some epidemiological data support the possibility that glucosinolate breakdown

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products derived from Brassica vegetables (cabbage, broccoli, and relatives) may protect against human cancers, especially in the gastrointestinal tract and lung.

ALDEHYDES AND KETONES

Aldehydes are any of a class of compounds characterized by the presence of a carbonyl group (C=O group) in which the carbon atom is bonded to at least one hydrogen atom. **Ketones**, on the other hand, are compounds where the carbon atom of the carbonyl group is bonded to two other carbon atoms. The citrus fruits, including orange (Citrus spp.), lemon (Citrus limon), as well as bergamot (Monarda didyma), may be cold-pressed to yield terpene-derived essential oils that are rich in aldehyde content, providing them with a unique aroma. The aldehyde and ketone components of these oils include **nootkatone**, **citral**, **octanal**, **sinensal.**

ESTERS

Esters are any class of compounds structurally related to carboxylic acids but in which the hydrogen atom in the carboxyl group (–COOH group) was replaced by a hydrocarbon group, resulting in a COOR structure (where R is the hydrocarbon). Thus, esters are formed through the condensation of alcohols (having an –OH group) and acids (having a –COOH group). They tend to have strong and often pleasant odors. Some of the volatile ester (and other) components present in strawberries (Fragaria chiloensis), apples (Malus spp.), and pineapples (Ananas comosus).

FATTY ACIDS

Fatty acids are the simplest lipids. They are characterized by a polar hydrophilic head region connected to a long hydrophobic tail. Some lipids, including the fats, are used for energy storage, but most are used to form **lipid/protein membranes.**

TERPENES

The **terpenes** have been prized for their essential oils and their use as fragrances for over two thousand years. Commerce in essential oils and **aromatherapy** continues today. For example, rose (Rosa spp.) fragrance has enchanted many. Bulgarian rose oil requires over 4000 kg of petals to produce 1 kg of steam-distilled oil. Over 260 constituents have been identified, many of which are olfactory relevant. The five compounds having the highest odor impact, listed in order of priority, are **damascenone**, **ionone**, **citronellol**, **cis-rose oxide**, and **phenethyl alcohol.**

ISOPRENE RULE

This rule describes all terpenes as having fundamental repeating five-carbon isoprene units. Thus, terpenes are defined as a unique group of hydrocarbon-based natural products that possess a structure that may be hypothetically derived from **isoprene**, giving rise to structures that may be divided into isopentane (2-methylbutane) units.

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Two different biosynthetic pathways produce the main terpene building block, **isopentenyl diphosphate** (IPP). The first is referred to as either the MEP (methylerythritolphosphate) or DOX (1-deoxy-D-xylulose) pathway. Here, IPP is formed in the chloroplast, mainly for the more volatile monoand diterpenes. The second biosynthetic route is known as the MVA (**mevalonic acid**) pathway. This takes place in the cytosol, producing sesquiterpenes. Terpenes are thus classified by the number of five-carbon units they contain:

- **Hemiterpenes:** C5: isoamyl alcohol, senecioic acid, tiglic acid, angelic acid, α- and βfuroic acid, and isovaleraldehyde
- **Monoterpenes:** C10: myrcene, geraniol, linalool, menthol, camphor, pinene, and limonene, thujone, Myrcene, Geraniol, Lavandulol, Perillene, Eucalyptol, camphor, Safranal
- **Sesquiterpenes**: C15: farnesol, artemisinin
- **Diterpenes:** C20
- **Sesterterpenes:** C25 (rare)
- **Triterpenes:** C30
- **Carotenoids:** C40

AROMATICS

Virtually all plants contain natural products that include a carbocyclic or heterocyclic aromatic ring thatgenerally contains one or more hydroxyl substituents. The vivid colors that light up the plants aroundus are generally composed from three sources: the tetrapyrroles, principally, chlorophyll; the terpenebased carotenes that we just discussed; and the aromatics. Several thousand aromatics are known, and new structures are continuously being discovered. In some cases, their functions are well known. For example, the polyphenolic lignins serve as structural components of the cell wall. In other cases, including the flavonoids, a variety of functions have been hypothesized, depending on the particular compound being investigated. Aromatic compounds are formed by several biosynthetic routes, including the polyketide and shikimate pathways, as well as from terpenoid origins. Due to the acidity of the phenol functionality (pKa of 8 to 11 depending on substituents), phenolic substances tend to have the potential for some water solubility and frequently form ether linkages with carbohydrate residues.

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BIOCHEMISTRY OF MAJOR METABOLIC PATHWAYS & PRODUCTS

PRIMARY METABOLIC PATHWAYS IN PLANTS

Some pathways are unique to plants, such as the carbon reduction cycle in photosynthesis and the **shikimic acid pathway** that produces, among other things, essential amino acids (like tryptophan) that animals cannot live without. These aromatic amino acids are also required for the production of many plant-specific nitrogen-containing and phenolic compounds. Microbes and mammals also have their own unique pathways, such as those involved in steroid hormone production, but common to plants, microbes, and mammals are the **pentose phosphate pathway**, **glycolysis**, and the **tricarboxylic acid(TCA)Cycle** that are concerned with aerobic respiration and **adenosine triphosphate (ATP)**biosynthesis— the key energy molecule of the cell.

CARBON ASSIMILATION: PHOTOSYNTHESIS

Photosynthesis converts atmospheric carbon dioxide into organic compounds that are either used in biosynthesis by the photosynthetic tissues themselves or converted to low-molecular-weight sugars (usually sucrose) and transported to nonphotosynthetic cells. Carbon dioxide is assimilated via a cycle of reactions that takes place in the stroma of the chloroplast. This process uses energy and reducing power supplied from light-driven reactions occurring in a specialized set of chloroplast membranes—the thylakoids. In this section we describe how physical energy (light) and inorganic carbon (CO_2) are captured and used by plants.

Net carbon assimilation occurs in the Calvin cycle

Carbon dioxide is assimilated through a cycle of reactions known as the reductive pentose phosphate pathway or the Calvin cycle (named after Melvin Calvin, who discovered it). This is the only pathway by which plants can convert a net amount of inorganic carbon into sugars. Its reactions are common to all autotrophic organisms and are the source of most of the organic compounds on earth.

The cycle reactions are catalyzed by soluble enzymes in the chloroplast stroma, and can be divided into three stages (1) carboxylation of a five-carbon compound (the CO2 “acceptor”) to give two molecules of a three-carbon compound; (2) reduction of the three-carbon compound to give triose phosphates; and (3) regeneration of the five-carbon acceptor from the triose phosphates.

The initial, carboxylation step is catalyzed by the enzyme Rubisco: carbon dioxide is combined with water and the five-carbon compound ribulose 1,5-bisphosphate to form two molecules of 3-phosphoglycerate. 3-Phosphoglycerate is then reduced to three-carbonsugars, the triose

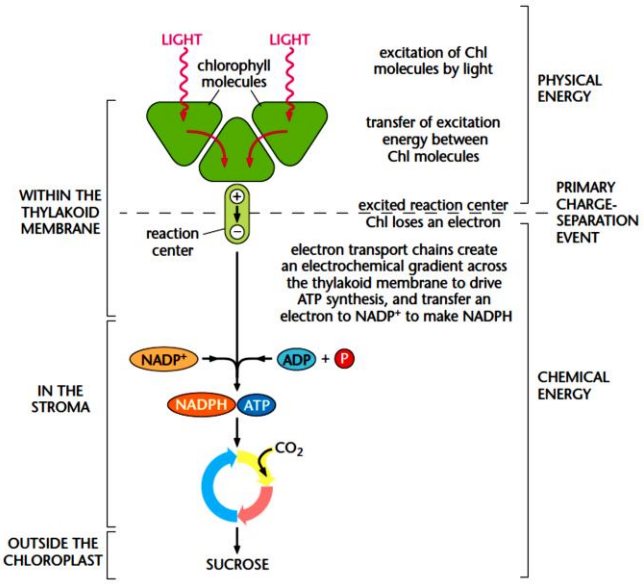
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phosphates glyceraldehyde 3-phosphate and dihydroxyacetone phosphate. The assimilation of three molecules of carbon dioxide in this cycle produces one more molecule of triose phosphate than is required for the regeneration of three molecules of ribulose 1,5-bisphosphate. Thus, for every five triose phosphate molecules used to maintain the supply of ribulose 1,5-bisphosphate for carboxylation, one can be exported from the chloroplast for the synthesis of sucrose. The cycle is "autocatalytic," in that it allows for removal of an intermediate for synthesis of net amounts of a product (sucrose) without compromising the rate of regeneration of the acceptor compound.

The reactions of the Calvin cycle require energy in the form of ATP and reductant in the form of NADPH (reduced nicotinamide adenine dinucleotide phosphate). The assimilation of 3 molecules of carbon dioxide—enabling the net synthesis of 1 molecule of triose phosphate for sucrose synthesis—uses 9 molecules of ATP and 6 molecules of NADPH. ATP and NADPH are synthesized in the thylakoid membranes, where light energy is captured and harnessed by electron transport chains.

Energy for carbon assimilation is generated by light-harvesting processes in the chloroplast thylakoids

The capture of energy from light and its use in the synthesis of ATP and NADPH for the Calvin cycle involves a complex series of physical and chemical reactions. Photons of light excite (energize) molecules of chlorophyll, a specialized pigment embedded in the thylakoid membranes. The energy of photoexcitation is then transferred by excited electrons to a reaction center, where this physical form of energy is converted to chemical energy by driving the transfer of one electron from a specific chlorophyll molecule to an acceptor molecule, thus leaving the chlorophyll molecule in an oxidized state ("oxidation" is loss of one or more electrons; "reduction" is the gain of electrons). This conversion of energy from a physical to a chemical state is known as the primary charge-separation event



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The reduced acceptor molecule in turn donates its newly received electron to an electron carrier, a molecule that accepts electrons then donates them to another molecule.

Several different electron carriers exist in the thylakoid membrane, and together these make up electron transport chains that transfer electrons from the acceptor molecules in the reaction centers to NADP+, reducing it to NADPH. The transfer of electrons along the chain drives the transport of protons across the thylakoid membrane from the stroma to the lumen (the internal space contained within the thylakoid membranes), setting up an electrochemical gradient across the membrane. The flux of protons back across the membrane from the lumen to the stroma, down the electrochemical gradient, drives the activity of membrane-bound ATP synthases, which phosphorylate ADP to ATP.

We will describe the stages of this process in detail: the initial capture of light energy and transfer of energy to the reaction centers; the transfer of electrons via electron transport chains to reduce NADP+ and set up a proton (pH) gradient; and the powering of ATP synthesis by the proton gradient.

Light energy is captured by chlorophyll molecules and transferred to reaction centers

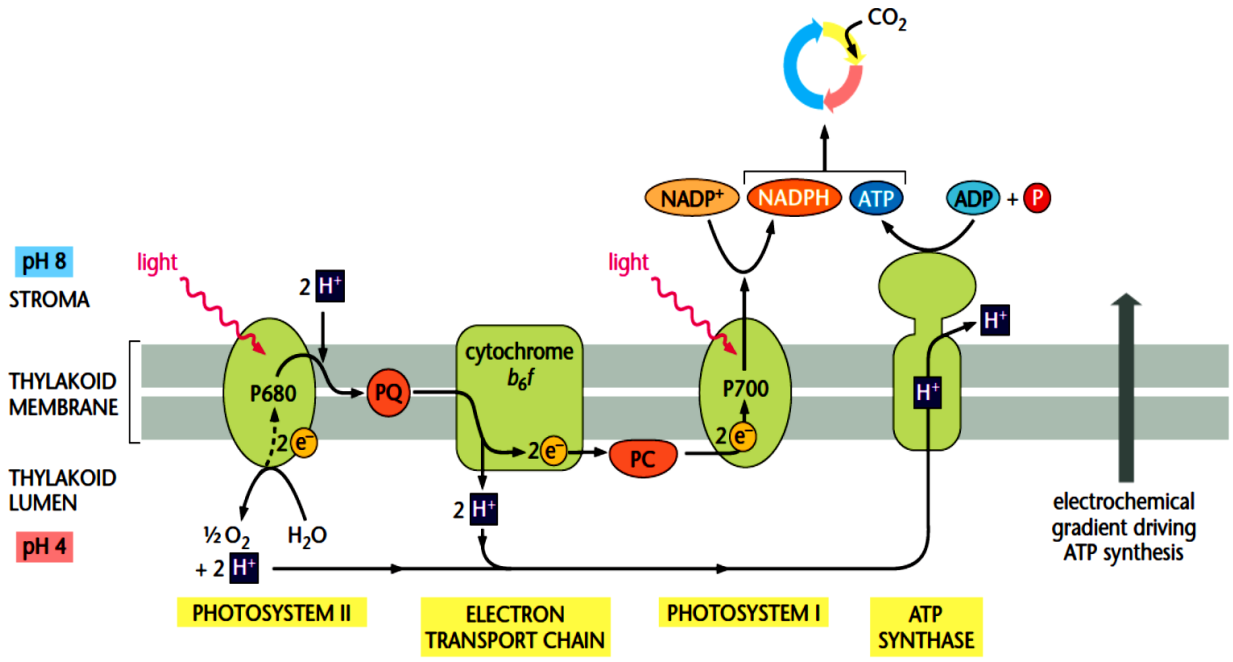
Light energy is captured by large, discrete complexes of chlorophyll molecules bound to proteins and embedded in the thylakoid membranes. Each of these discrete complexes constitutes a photosystem, each containing about 250 chlorophyll molecules. The majority of the chlorophyll molecules in a photosystem make up the antenna. The antenna molecules channel excitation energy toward the remaining chlorophyll molecules, which make up the core complex. The reaction center is contained within the core complex, and it is here that light energy is transduced into chemical energy. In the antenna, chlorophyll is bound to proteins known as "light-harvesting complex polypeptides" (LHCPs); these pigment–protein complexes span the thylakoid membrane, with chlorophyll molecules positioned to maximize the efficiency of energy transfer among them. The light-harvesting complexes contain two types of chlorophyll molecule, chlorophyll a and chlorophyll b, in an average ratio of about 3 to 1. The complexes also contain related pigment molecules known collectively as carotenoids.

When antennal chlorophyll absorbs a photon, energy is transferred to adjacent chlorophyll molecules and then channeled toward the reaction center in the photosystem. Chlorophylls a and b have different absorption spectra: chlorophyll a is excited by photons of lower energy (longer wavelengths of light) than chlorophyll b, and thus achieves a lower energy state than b. This means that excitation energy passes readily from chlorophyll b to chlorophyll a, but not in the other direction. The ratio of chlorophyll a to b increases toward the reaction center, thus channeling excitation energy toward the reaction center. In addition, chlorophyll a molecules associated with the reaction centers are excited by photons of lower energy (i.e., longer wavelength) than antennal chlorophyll a—similarly promoting the transfer of excitation energy toward the reaction center. The absorption spectra of antennal and reaction- center chlorophyll a differ because the chlorophylls in the two locations are bound to different types of protein.

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Electron transfer between two reaction centers via an electron transport chain reduces NADP+ and generates a proton gradient across the thylakoid membrane.

Higher plants contain two types of photosystem: photosystem I (PSI) and photosystem II (PSII); the two work in series, connected by an electron transport chain. In essence, primary charge-separation events in the PSI reaction center cause the reaction-center chlorophyll a molecule to donate electrons to NADP+ via an electron transport chain, thus making the NADPH required by the Calvin cycle. The oxidized chlorophyll a molecule in the PSI reaction center is then re-reduced by electrons generated by primary charge-separation events in PSII and transferred to PSI by an electron transport chain. In turn, the oxidized chlorophyll a in the PSII reaction center is re-reduced by electrons donated from water, via a water-splitting complex that is an integral part of the PSII reaction center. The photosystems and the components of the electron transport chain are asymmetrical complexes, organized across the thylakoid membrane in such a way that reductive reactions (in which protons are taken up) occur on the stromal side of the membrane, and oxidative reactions (in which protons are released) occur on the lumenal side of the membrane. This leads to a net transfer of protons from the stroma to the lumen, setting up both a proton (pH) gradient and an electrical potential across the membrane. This electrochemical gradient drives the synthesis of ATP by the ATP synthase complex embedded in the membrane. We consider first the structure and operation of the photosystems and the electron transport chain that links them.



The proton gradient drives the synthesis of ATP by an ATP synthase complex

The way in which electron transport chains in the membranes of chloroplasts and mitochondria drive ATP synthesis was first explained by Peter Mitchell in the 1960s. In his chemiosmotic model, Mitchell proposed that electron-transfer processes in the membrane result in a net movement of protons from one side of the membrane to the other, generating both a chemical gradient (a gradient of proton concentration) and an electrical potential across the membrane. Mitchell called this electrochemical gradient the proton-motive force. The energy released by dissipation of the proton-motive force— by passage of protons down the electrochemical gradient through a channel in a membrane- spanning complex—is used to drive the synthesis of ATP. In chloroplasts thisprocess is called photophosphorylation.

In chloroplasts, a net transfer of protons across the thylakoid membrane, from the stroma to the lumen, takes place at two points in the transfer of electrons from water to NADP+. First, oxidation of water on the lumenal side of the membrane frees protons into the lumen. Second, electron transfer to and from plastoquinone results in the transfer of protons from the stroma to the lumen. The release of protons at these two points produces a large gradient of protons and a small electrical potential across the thylakoid membrane. In conditions favorable for photosynthesis, the difference in proton concentration between the lumen and the stroma may be as much as four orders of magnitude (4 pH units): the pH values of the lumen and the stroma are about 4 and 8, respectively.

The ATP synthase complex is located in the stromal thylakoid membrane. Its three dimensional structure is well understood. The complex consists of two major components: the transmembrane portion (the CF0 segment), which contains the channel for proton movement, and the portion that projects into the stroma (the CF1 segment), which synthesizes ATP. The CF0 segment consists of four types of protein, three chloroplast-encoded and one nuclear-encoded. The CF1 segment consists of three chloroplast-encoded and two nuclear-encoded proteins.

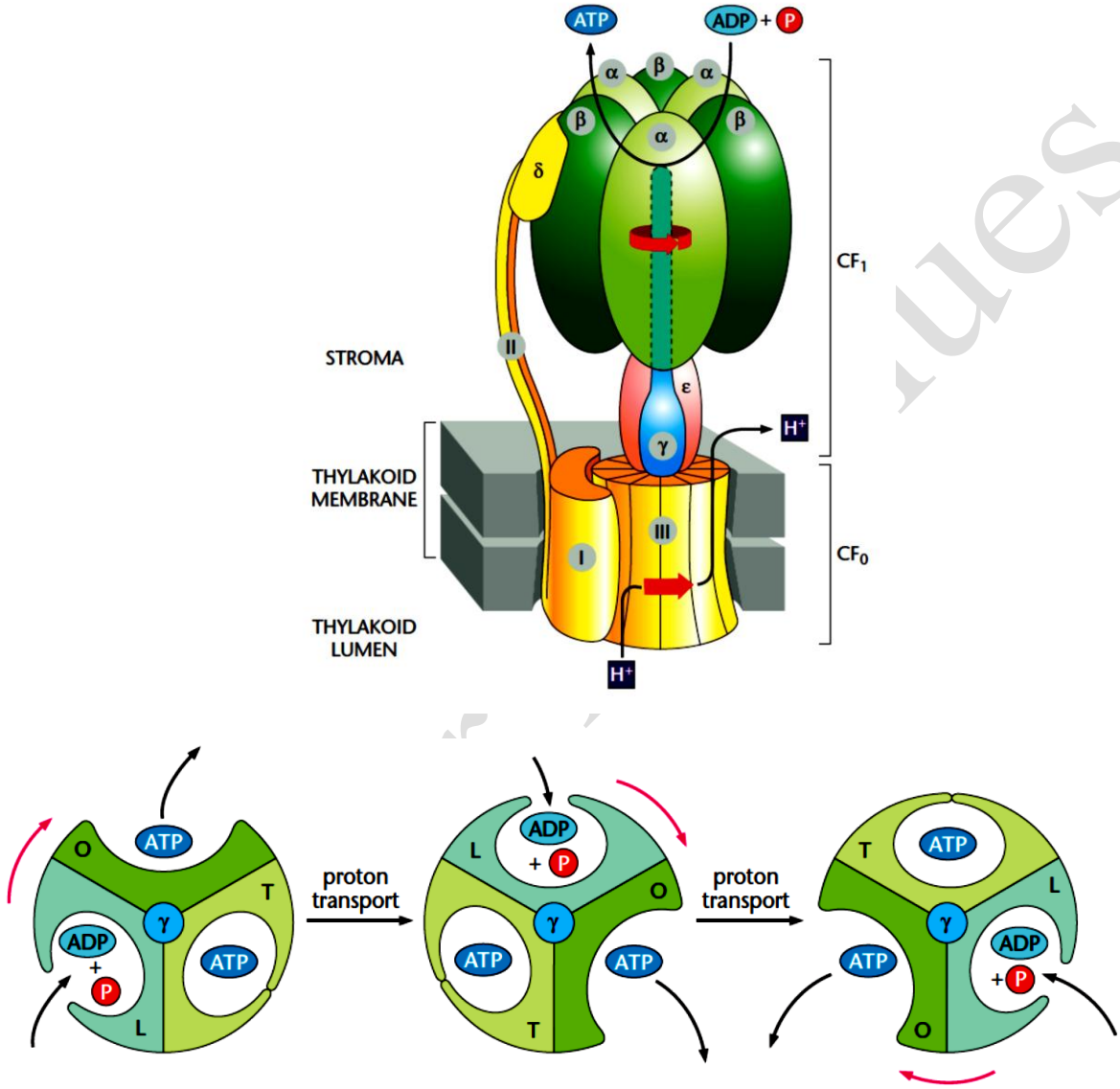
How the passage of protons through CF0 is coupled to the synthesis of ATP by CF1 is not clear. It is generally agreed, though, that the flux of protons brings about a cycle of conformational changes in the catalytic sites of CF1, and these changes drive ATP synthesis.

At a given time, the three catalytic sites of CF1 exist in three different states: loose, tight, and open. Conformational changes brought about by proton flux through CF0 drive the three sites through a cycle of all three states. In the loose state, a site binds the substrates for ATP synthesis, ADP and inorganic phosphate; conversion to the tight state promotes the synthesis of ATP; and finally, conversion to the open state releases the ATP.

The ATP synthase complex is strictly regulated by light and by the pH gradient across the thylakoid, such that it is inactive in the dark and when rates of electron transport are low. This regulation is important because the ATP synthase activity is potentially reversible—if active in the absence of a pH gradient sufficient to drive ATP synthesis, it could hydrolyze stromal ATP and drive proton movement from the stroma to the lumen (i.e., acting as an ATPase). The catalytic sites of CF1 are inactivated when the pH gradient drops below a certain level. Their

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activity also depends on the reduction of a disulfide bond on one of the CF1 subunits. The reduction is brought about by reduced thioredoxin, an important sensor of the rate of electron transport: thioredoxin is reduced in the light and oxidized in the dark, hence CF1 is active only in the light



Carbon assimilation and energy supply are coordinated by complex regulation of Calvin cycle enzymes

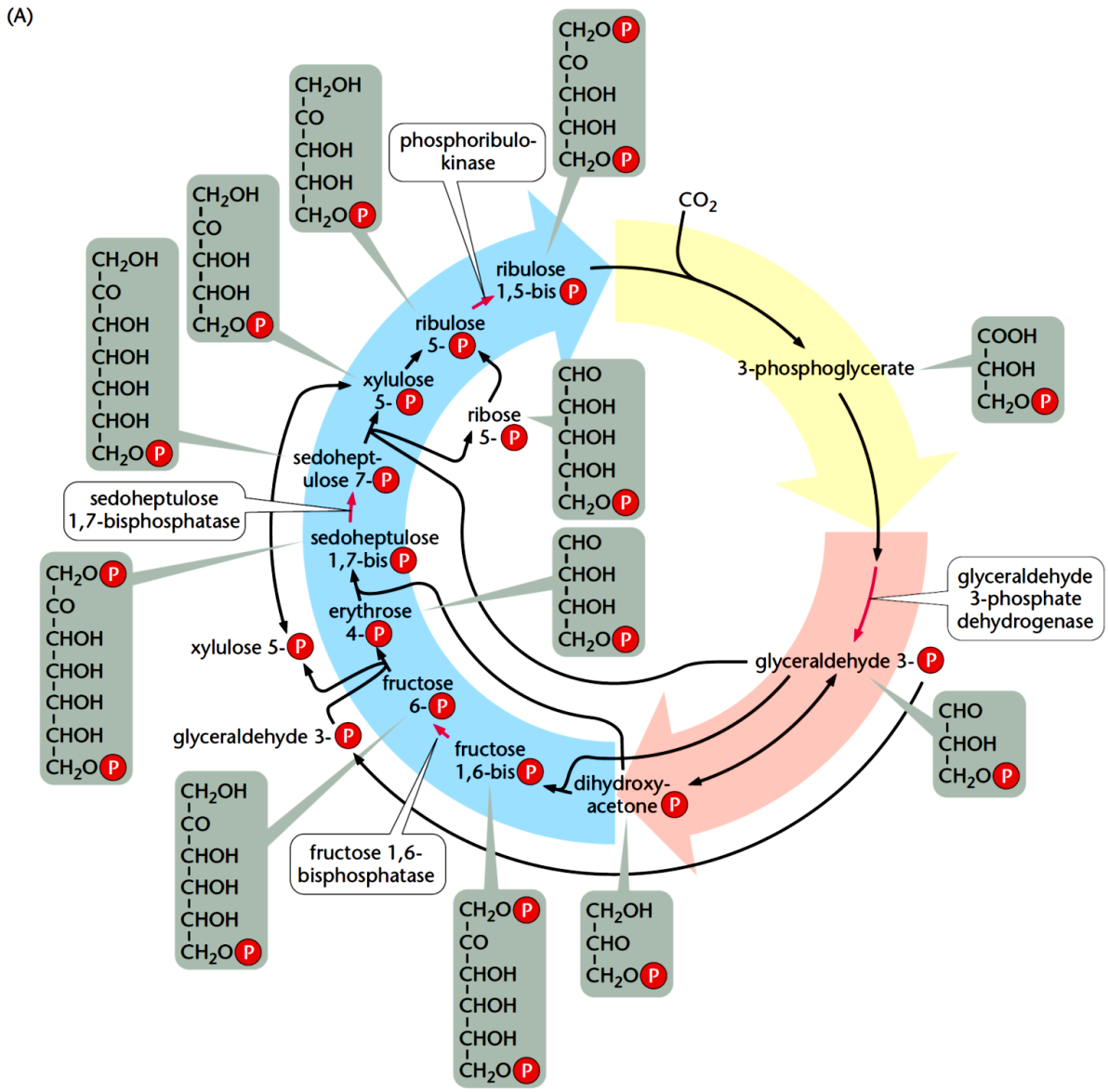
Integration of Calvin cycle activity with the supply of ATP and NADPH is essential for a plant's efficient and sustained assimilation of carbon dioxide. When energy supply changes due to changing environmental conditions, coordinated regulation of certain Calvin cycle enzymes ensures that the pools of intermediates are maintained at optimal levels, thus allowing sensitive modulation of the balance between energy supply and carbon assimilation.

The main mechanism by which Calvin cycle enzymes are regulated by the cell's energy supply is reductive activation. Some enzymes of the cycle are active when the sulfhydryl groups of particular cysteine residues are in a reduced state (–SH) and inactive when these groups are oxidized as disulfide bonds (S–S). The reduction state, and hence the level of activation, is controlled by a system that responds directly to the state of reduction of ferredoxin, the electron-carrier protein of PSI that donates electrons to NADP+. Besides reducing NADP+, reduced ferredoxin also reduces thioredoxin, in a reaction catalyzed by ferredoxin-thioredoxin reductase. Reduced thio redoxin in turn reduces the disulfide bonds of some Calvin cycle enzymes, thus activating them. Reduced thioredoxin can also be oxidized by oxygen, and the balance between the oxidized and reduced states depends on the state of reduction of ferredoxin, and hence on the flux of electrons through the photosystems. In its oxidized state, thioredoxin oxidizes the sulfhydryl groups and thus deactivates the Calvin cycle enzymes. In this way, the level of activation of several enzymes is directly linked to the flux of electrons through the photosystems and hence to the availability of NADPH and ATP for Calvin cycle reactions.

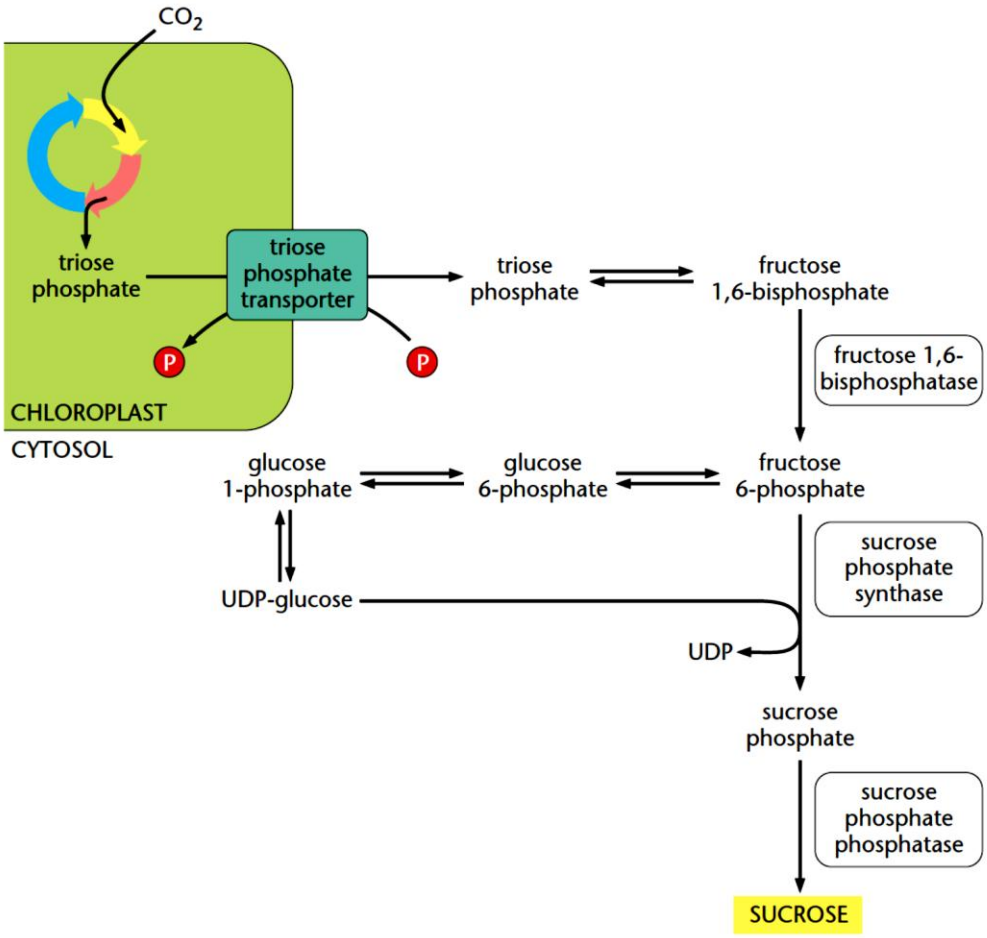
The level of activation of Rubisco, the carbon dioxide–assimilating enzyme , is also linked to light-driven reactions, but by a different mechanism that involves the enzyme Rubisco activase (Figure 4–23). Rubisco activase is activated by the reduction state of PSI, probably via the ferredoxin-thioredoxin system in the same manner as the Calvin cycle enzymes described above. When active, Rubisco activase promotes the binding of carbon dioxide to a specific lysine residue in the active site of Rubisco, located on the large subunits of the enzyme. This carbamylation reaction promotes Rubisco activity.

The light-driven reactions of photosynthesis also regulate the activity of Calvin cycle enzymes in more generalized ways. The pH of the stroma (where the Calvin cycle reactions occur) is strongly influenced by the electron-transfer processes in the thylakoid membranes. As described above, the generation of the proton gradient that drives ATP synthesis removes protons from (increases the pH of) the stroma. This proton movement results in the movement of other cations—notably Mg^{2+}—from the thylakoid lumen into the stroma. Thus the stroma is at about pH 7 with 1 to 3 mM Mg^{2+} at night, but about pH 8 with 3 to 6 mM Mg^{2+} in the light, when electron transport is occurring. The higher pH and Mg^{2+} concentration increase the activation state of several Calvin cycle enzymes, including Rubisco (these conditions promote activation by favoring carbamylation). Activity of several Calvin cycle enzymes is also modulated by intermediates of the Calvin cycle itself. For example, Rubisco is inhibited by a buildup of its product, 3-phosphoglycerate.

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The Calvin cycle



Synthesis of sucrose in the cytosol. Triose phosphate generated in the Calvin cycle is transported to the cytosol, where it undergoes a series of conversions involving six-carbon sugars—as sugar phosphates or sugar nucleotides—followed by joining of a glucose and a fructose residue to yield the disaccharide sucrose. Four triose phosphate molecules yield one sucrose molecule.

LIPIDS, PROTEINS, AND NUCLEOTIDES

All living organisms produce three major categories of compounds:

- **Lipids** that make up the plasmamembrane and the membranes of all internal compartments and organelles;
- **Proteins** that make upstructural units of the cell, such as microtubules, and all the enzymes of every biochemical process; and
- **Nucleic acids** and **nucleotides** that code for all proteins, act as metabolic energy molecules such as ATP and biochemical regulators such as GTP or cAMP, and in some cases, work in conjunction with proteins to produce specific activities.

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Ribosomes, for example, consist of both protein and RNA, the combination of which allows for the production of all other proteins. Because all organisms produce these compounds.

Lipids are highly hydrophobic compounds produced by a partnership between plastids and the ER. Most lipids have a fatty acid portion made from acetyl-CoA and malonyl-CoA in areaction that produces longer molecules with repetition. Malonyl-CoA is simply the carboxylated form of acetyl-CoA. In animals, fatty acid biosynthesis takes place in the cytosol, but in plants, it occurs in plastids (chloroplasts in green tissue; proplastids in nongreen tissue). In higher plants and animals, the predominant fatty acid residues are those of the C16and C species of palmitic, oleic, linoleic, andstearic acids. However, there are many different forms of lipids. Membrane lipids such asphospholipids and glycolipids are made from a combination of glycerol, fatty acids, and hydrophilic compounds, such as serine, choline, inositol, or various sugars. The many varieties of phospholipids and glycolipids are made from phosphatidate, a phosphorylated sugar derivative that acts as the precursor for the polar heads of these lipids. Vesicles that bud off of the ER or Golgi apparatus carry specific phospholipids to their proper locations in the plasma membrane or organelles. In addition to the typicallipid cell components, plants have different metabolic pathways that produce **waxes** that make up the protective cuticle of epidermal cells, and **terpenes** that are lipids synthesized from acetyl CoA via the mevalonic acid pathway in the cytosol as well as from pyruvic acid and glyceraldehyde-3-phosphate via 1-deoxy-D-xylulose-5-phosphate (DOXP) and 2-C-methyl Derythritol-4-phosphate (MEP) in the plastids. Terpenes produced in the terpenoid pathway serve a huge variety of function sin photosynthesis, hormone-controlled development (gibberellin and abscisic acid), and flower coloration and scent to name a few. For humans, they are a source for rubber, essential oils (perfumes), and medicinal drugs such as Taxol (an anticancer drug). Plants produce very important storage forms of lipids (fats and oils) as energy reserves in fruits and seeds, such as the fats and oils found in avocados, olives, soybeans, sunflower seeds, and peanuts. In some cases, the sereserves may also serve as rewards for animals that disperse the plant's seeds. These stored lipids are often found in the cytoplasm of either cotyledon or endosperm cells in organelles known as spherosomes (also called lipid bodies), which, like vesicles, bud off of the ER.

The production of proteins is completely dependent on the presence of the amino acids from which they are made as well as nucleotides, because every protein is coded by nucleic acids (DNA and RNA) made from nucleotides (see Chapter 1 for more details). In eukaryotic cells, most proteins are initially produced in the cytosol and then transported to their final destinations in the cells where they will perform their specific functions. Organelles, such as chloroplasts and mitochondria, can also make proteins specific to these organelles. We mentioned that proteins may be enzymatic or structural in function, but plants produce storage forms of proteins, like **phytate**, to provide a reserve of amino acids and energy, especially in the process of seed germination. Some of these storage proteins can be **lectins**, which are highly toxic and serve as herbivore deterrents, but their ability to bind sugars gives them function in recognition of symbionts, pathogens, and species-specific pollen grains as well.

The purine and pyrimidine nucleotides that allow the synthesis of nucleic acids (DNA or RNA) are made in the cytoplasm from sugars and aliphatic amino acids. Purine nucleotides are made

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from ribose-5-phosphate, a modified ribose sugar, while pyrimidine nucleotides also require glutamine. So, a nucleotide is simply one of several different nitrogen-containing ring compounds linked to a five-carbon sugar (either ribose or deoxyribose) that carries a phosphate group (see Chapter 1 for more details). Nucleotides are also salvaged within the cell from the degradation or breakdown of nucleic acids (usually RNA).Please remember that all biochemical processes are ultimately controlled by the timing of the expression of the genes encoded by DNA.

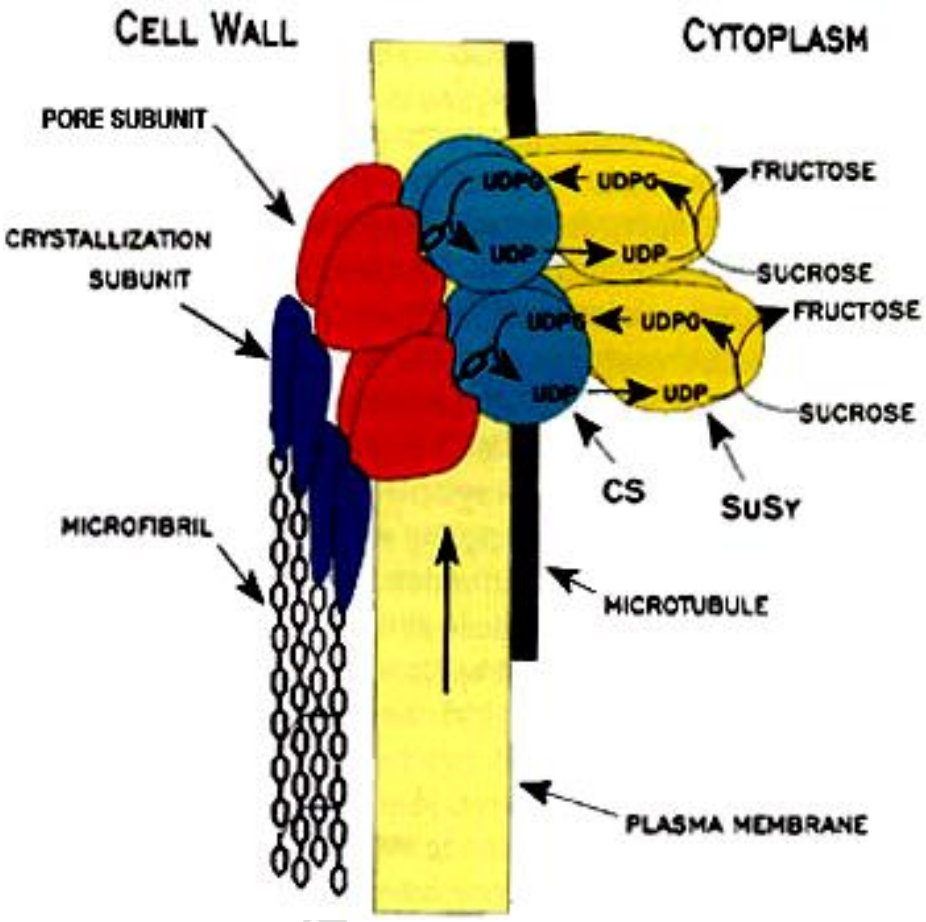
CELLULOSE AND CELLULOSE BIOSYNTHESIS

Celluloseis is the world's most common naturally synthesized polymer. It makes up the majority of all the biomass on the planet and is the primary component of all plant cell walls. This homopolymer is made from the glucose molecules produced by photosynthesis and is organized as glucanchains of β-1, 4-linked glucose units in which every other glucose unit is rotated 180^0 with respect to its neighbor. The glucan chains in primary walls of growing plant cells aggregate into fibers called **cellulose microfibrils**. In secondary walls, laid down after cell growth has ceased, the cellulose microfibrils are organized into macrofibrils or bundles. Cells that expand more or less equally in all directions have cellulose microfibrils oriented in a random pattern; in contrast, cells that expand by elongation growth (e.g., fibers, pollen tubes, root hairs, and conducting cells of the vascular system) have cellulose microfibrils oriented parallel to each other, lying at right angles to the direction in which the cell elongates. These patterns of orientation of cellulose microfibrils help govern the specific function of a given cell and can be determined microscopically by the use of crossed polarizers and a red filter placed diagonally to the crossed polarizers.

The synthesis of cellulose occurs at the plasma membrane, which is located at the interface between the cell wall and the cytoplasm. The monomeric unit that donates glucose units to a growing cellulosechain is **UDPG (uridine diphosphate glucose)**. The glucose in UDPG comes from the hydrolysis of the disaccharide sugar, sucrose, catalyzed by the enzyme, **SuSy (sucrose phosphate synthase)**.

The cell wall in plants provides structural support for the plant. This structural support is provided not only by cellulose, but also, by other polymers, such as hemicellulose and pectic polysaccharides. Like cellulose, these are chains of sugars, but their many varieties differ from cellulose in the kinds of sugars present, how they are linked together, and how many branches they have in their chains. Hemicellulose and pectins are not made at the plasma membrane. Instead, they appear to be made in the secretory system of the ER and Golgi apparatus. They are then transported to the cell wall via vesicles. In woody plants, such as vines, trees, and shrubs, the cell walls become lignified through deposition of the polymer, **lignin**. Cellulose combined with lignin is the primary plant product involved in support and provides the physical structure that allows such plants as trees to grow very tall. In vascular plants, such as grasses, sedges, and scouring rushes (Equisetum spp.), as well as in diatoms (one type of algae), the cell walls become infiltrated with **amorphous silica gel** that, like lignin, provides structural support.

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LIGNIN AND LIGNIN BIOSYNTHESIS

Lignin is a complex polymer that exists as a three-dimensional matrix around the polysaccharides of secondary cell walls found in plant fibers and in the tracheids and vessel elements of secondary xylem (wood). It is composed of varying amounts of the aromatic phenylpropanoid subunits (monolignols), p-coumaryl alcohol, coniferyl alcohol, and sinapyl alcohol made via the shikimic acid pathway. These monolignols are usually synthesized from the amino acid L-phenylalanine, although tyrosine can also be used. These monolignols appear to be made in the ER and Golgi bodies, but the polymerization of lignin occurs outside the plasma membrane.

BIOGENIC SILICA AND SILICIFICATION

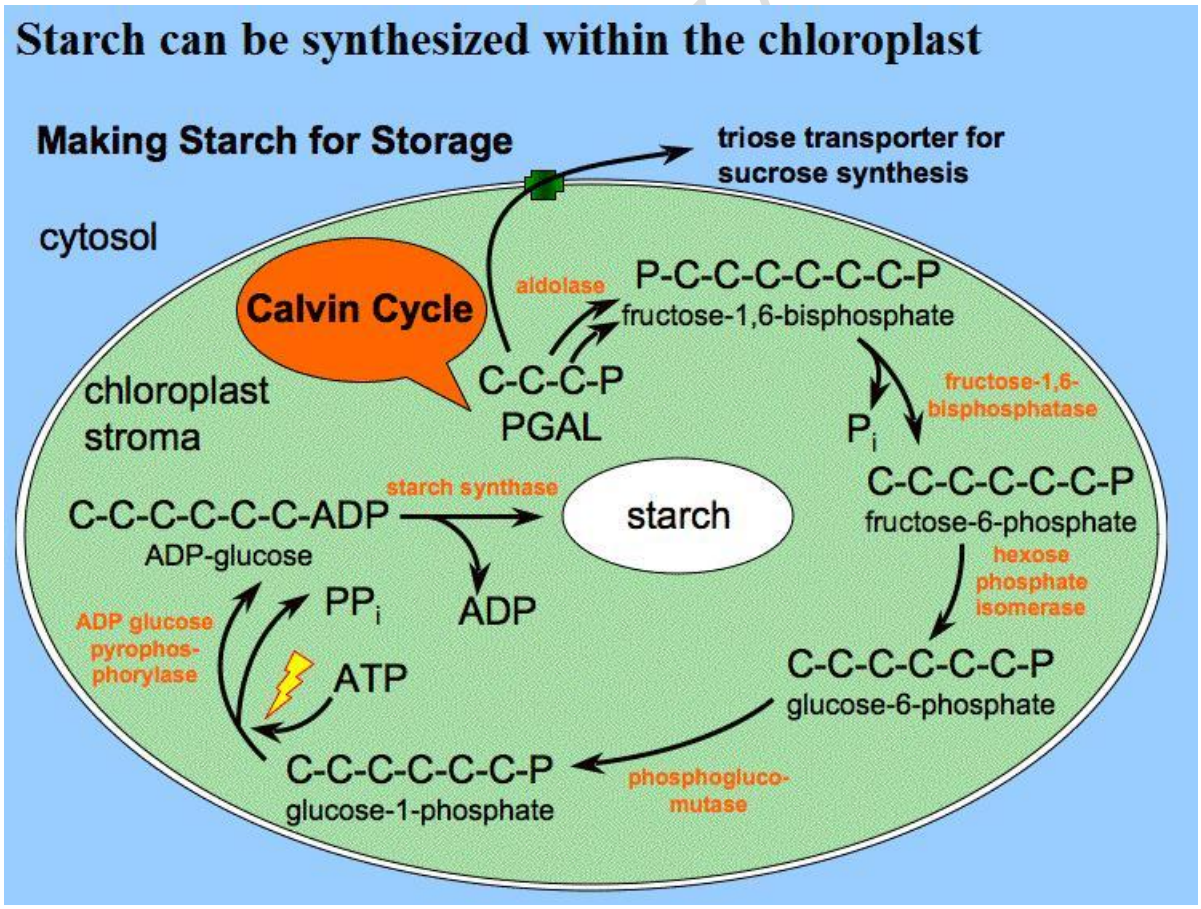
Some plants have developed the ability to absorb inorganic constituents from their environment and use them for their benefit. Biogenic silica is a polymer of biological origin that is characteristically found in the cell walls of diatoms, scouring rushes (Equisetum spp.) or horsetails, grasses (all members of the Poaceae or grass family), members of the rush family

(Juncaecae), and members of the sedge family (Cyperaceae). Silica found in these silica-accumulating plants originates from silicates found in soil minerals. It is taken up as monosilicic acid, Si(OH)4, via the roots (or cell membrane in the case of the single-cell diatoms), from which it moves up the plant in the xylem-conducting elements.

STARCH AND STARCH BIOSYNTHESIS

Starch is the most common storage polysaccharide found in plants, and it serves as a primary food source for humans, domestic animals, birds, insects, and microbes. **Starch** is essentially comprised of monomers of sugars linked end to end in long chains through α-1,4 linkages along the chain with α-1,6 linkages as branch points.

The other configuration of the starch molecule is that of **amylose**, which is unbranched. It is composed exclusively of α-1,4-linked glucan. In rice grains, for example, varying amounts of **amylose** (straight-chain starch) and **amylopectin** (branched-chain starch) are found, depending on the cultivar. In sticky, short-grain rice (Japonica cultivars), amylopectin predominates. Such rice is better for soups and for eating with chopsticks. In non-sticky, long-grain rice (Indica cultivars), amylose starch predominates. Both forms of starch are made in the endosperm tissue of the seeds.



FRUCTANS AND FRUCTAN BIOSYNTHESIS

Fructans are soluble storage polysaccharides found in the vacuoles of cells of plants known to be fructan accumulators. They are made predominantly from the sugar **fructose** (hence, the name), but glucose sugars may also be present in the chain. Classic examples include temperate-zone monocots, such as grasses, lilies, and irises, and dicots, such as dahlias and Jerusalem artichokes. In dahlia and Jerusalem artichoke, the fructan is referred to as **inulin**, and here, the polysaccharide is composed mostly or exclusively of 2–1 fructosyl–fructose linkages (glucose is allowed but not necessary). In the temperate-zone grasses, the fructan is termed either **graminan**, which has both 2–1and 2–6 fructosyl–fructose linkages, or **phlein**, which contain mostly or exclusively 2–6 fructosy l fructoselinkages. As with inulin, glucose is allowed in the chain but is not necessary in the structure of graminan- and phlein-type fructans.

GUM, MUCILAGE, AND DIETARY FIBER

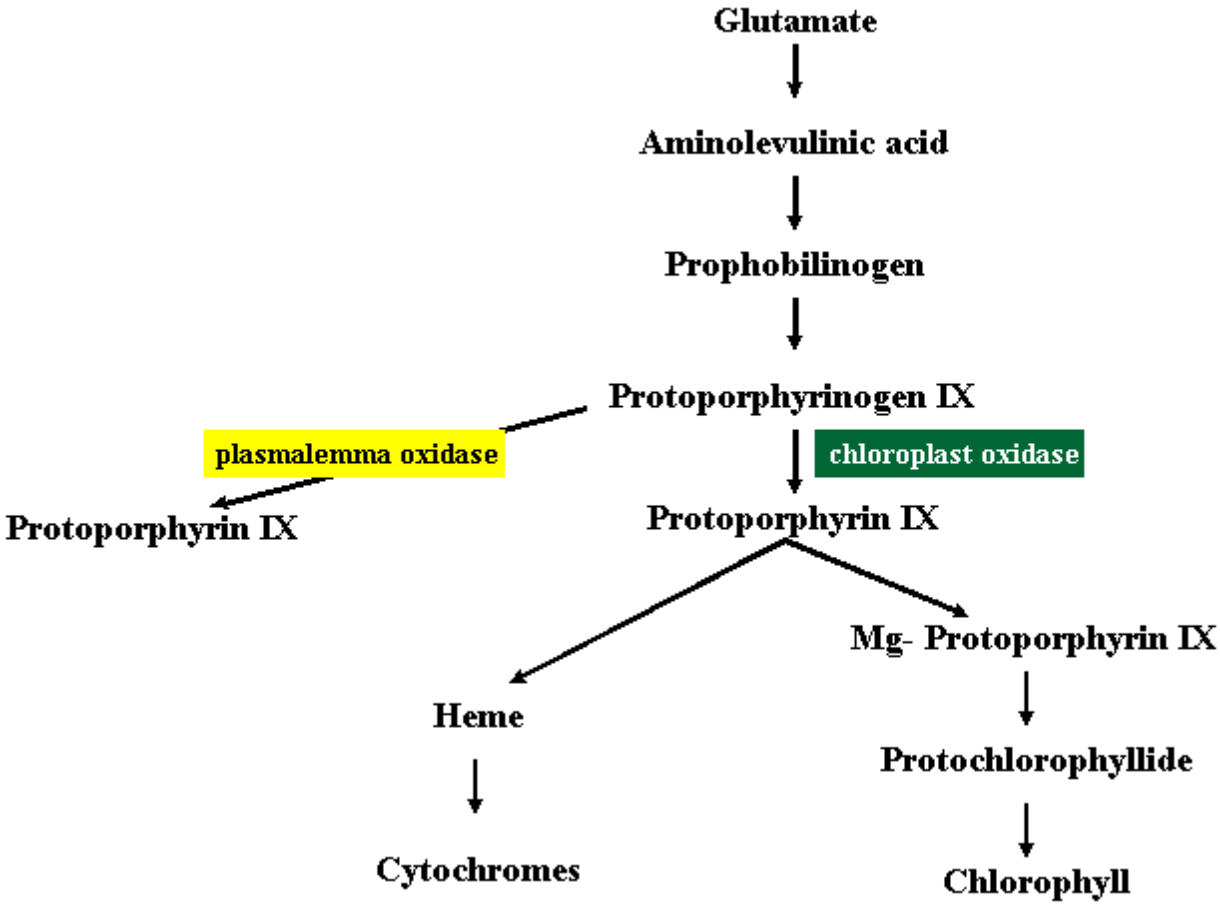
Plants produce other forms of polysaccharides that include **gum** and **mucilage**. These are highly branched heteropolysaccharides (related to hemicellulose and pectic polysaccharides) that contain acidic residues, thus making them very hydrophobic, insoluble within the plant cell, and often difficult for animals to digest. One benefit to the plant that produces indigestible polymers is that it reduces the reward for herbivores. In other words, the animal spends its time eating, yet gets nothing out of the process. For the plant, these polysaccharides can function as a storage reserve for carbohydrates, but they are also found as part of the matrix that surrounds the cell walls of some cells. This matrix is called the **glycocalyx** and is mostly seen on the surfaces of roots, where it may serve to protect the plant against microbial invasion. Glycocalyx secretions are not unique to plants. They are also found in bacteria and in animals, where, as in plants, they act in cell–cell recognition of symbionts or pathogens. Another function of mucilage is seen in carnivorous plants like sundew (Drosera rotundifolia), where a substance called **mucin** is produced to catch unwary insects in nutrient-poor environments. Gums are also useful in the sealing of wounds in leaves and stems. For example, when a cherry tree is injured, it will produce a thick substance called **gum arabic** that fills in the wound, thus preventing infection. This also acts as a human cosmetic.

CHLOROPHYLL AND CHLOROPHYLL BIOSYNTHESIS

There are three main locations of pigments within the cell: (1) plastids, (2) vacuoles, and (3) cell walls. The chemistry of the pigments varies with the location. **Chlorophyll** is a porphyrin that constitutes the primary photoreceptor pigment for the process of photosynthesis in plants. It is produced in the chloroplastsand is responsible for the green appearance of leaves and stems, aerial and prop roots, many kinds of floral bracts, and green fruits before they ripen. The chlorophyll molecule is made up of four pyrrole rings, made from aliphatic amino acids (designated I to IV), that are ligated to form a **tetrapyrrole**ring with a magnesium atom in its center . Ring IV is esterified with a hydrophobiclong-chain **phytol** molecule (C20H39) made in the terpenoid pathway. For light harvesting, plants use two forms of chlorophyll — ‘a’ and ‘b’. Chlorophyll a is in all plants and is the only chlorophyll at the reaction centers. It has a methyl group at C3. Chlorophyll b, found in most plants, has a formyl group at this position and, like

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other accessory pigments, functions to absorb the energy from wavelengths of light that differ from chlorophyll a.



CAROTENOIDS AND CAROTENOID BIOSYNTHESIS

Plant carotenoids are responsible for the red, orange, and yellow pigments found in fruits and roots, including tomatoes, red peppers, pumpkins, and carrots. They can be seen in the petals of many flowers and are the primary pigments responsible for the fall coloration of deciduous trees. Carotenoids are synthesized in the terpenoid pathway as C40 tetraterpenes derived from the condensation of eight isoprene units starting with isopentenyl diphosphate (Figure 2.18; Bartley and Scolnik, 1995; Britton, 1995; McGarvey and Croteau, 1995; Dey and Harborne, 1997).

There are two basic types of carotenoids: (1) **carotenes** that contain no oxygen atoms and (2) **xanthophylls** that contain oxygen. At the center of each carotenoid molecule, the linkage order is reversed, resulting in a molecule that is symmetrical. A set of double bonds in the molecule is responsible for the absorption of light in the visible portion of the spectrum. As mentioned above, this has an important impact on the absorption of a wider range of light wavelengths for use in photosynthesis. Consequently, in photosynthetic organisms, carotenoids are an integral structural component of photosynthetic antenna and reaction center complexes, but they also protect against the harmful effects of photooxidation processes. Like chlorophyll,

carotenoids are found in the thylakoids of green leaves and stems. In fruits and flowers, they are also found in plastids, but these plastids have structural differences and are referred to as **chromoplasts** to indicate that they contain pigments other than chlorophyll. **β-carotene** is the orange pigment in carrot (Daucus carota) roots, sweet potato (Ipomoea batatus) tuberous roots, pumpkin (Cucurbita Pepo) fruits, leaves of deciduous trees, and some flower petals.

Zeaxanthin and violaxanthin are found in autumn-colored leaves and flower petals and are responsible for the bright yellows that are sometimes seen. Coloration of flowers is very important to the survival success of the plants producing them. The color of the flowers is one of the primary factors involved in attracting pollinators. For more information on the attraction of pollinators, β carotene is important in the human diet because of its purported anticancer activity, its use as a food coloring, and it is an important source of **vitamin A** (an alicyclic alcohol), which is synthesized from β carotene and other carotenoids. Vitamin A produced by animals is, in turn, converted to the pigment, **retinal.**

In ripening fruits, as in leaves turning color in autumn, chlorophyll pigments gradually break down in chloroplast thylakoid membranes, revealing the carotenoid pigments that were masked by the chlorophyll pigments. During ripening, there is significant synthesis of new carotenoid pigments. In the case of ripening tomatoes and peppers, for example, the unripe fruits are characteristically bright green. As ripening progresses (triggered by the plant hormone, **ethylene**), various carotenoid pigments appear and, newly synthesized, account for the color of the ripe fruits. **Lycopene** is the red pigment seen in mature tomato and red pepper fruits. Tomatoes (Lycopersicon esculentum) can have both red and yellow fruits depending on the genotype of the parent. In some peppers, ripe fruits are green (sold as sweet bell peppers) but turn red or yellow at maturity (Capsicum frutescens var, grossum). Other ripe peppers, often very "hot" to the taste ("hot" due to the presence of the alkaloid, **capsaicin**), may be green, orange, yellow, or red at maturity, depending again on the genotype of the parents. Similar types of color changes occur in ripening cucurbit fruits (squash, gourds, and pumpkins in the Cucurbitaceae family) and in the fruits of eggplant (Solanum melongena var, esculentum).

ANTHOCYANINS AND ANTHOCYANIN BIOSYNTHESIS

Anthocyanins are flavonoid-type compounds responsible for most of the red, pink, purple, and blue pigments found in roots, stems, leaves, flowers, seeds, and fruits (Williams and Grayer, 2004). Examples include the red anthocyanins in red radish (Raphanus sativa), the red leaves of some Norway maplecultivars (e.g., Acer saccharum cv. 'Schwedleri'), the red fruits of some peppers (Capsicum frutesens), apples (Malus sylvestris), and Acerola cherry (Malpighia glabra, said also to contain the highest content of vitamin C, ascorbic acid, of any fruit), and the red, pink, purple, and blue flowers of Rhododendron, Hibiscus, and Fuchsia, to name a few. Anthocyanin pigments occur in the vacuoles of plant cells. They are synthesized from the aromatic amino acid, phenylalanine, in the phenylpropanoid pathway. This is the same pathway responsible for the synthesis of tannins; flavan derivatives; isoflavones like genistein, daidzein, and pterocarpans; lignin;lignans; and coumarin. The primary enzyme that commits the pathway

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to biosynthesis of the anthocyanin pigments is **chalconesynthase** (**CHS**). There is a whole gene family of CHS genes within most plants. Some of the genesare expressed in specific tissues.

Anthocyanins serve many diverse functions in plants, including attraction of insect and bird pollinators to flowers and dispersal of seeds and fruits by birds and mammals. In some cases, they are feeding deterrents, and like other flavonoids, they can protect the plant against damage from UV irradiation. Anthocyanins have great economic importance in expression of the wide array of flower colors inplants grown as ornamentals. In fact, attempts to obtain blue roses, chrysanthemums, and carnations are now possible with transgenic plants. In these plants, synthesis of the blue pigment, delphinidin-3-glucoside, does not normally occur, because the 3,5-hydroxylase, is not normally expressed. In the transgenic plants, this gene, obtained from other plants like petunia, is expressed, resulting in the synthesis of the blue delphinidin-3-glucoside anthocyanoid pigment. One interesting application in the use of naturally occurring anthocyanin pigments comes from the pigment present in the red roots of radish, Raphanus sativus. This water-soluble pigment is extracted from these roots and is currently used to dye, Marashino cherries bright red instead of using a synthetic red dye as was done previously.

ALKALOIDS AND ALKALOID BIOSYNTHESIS

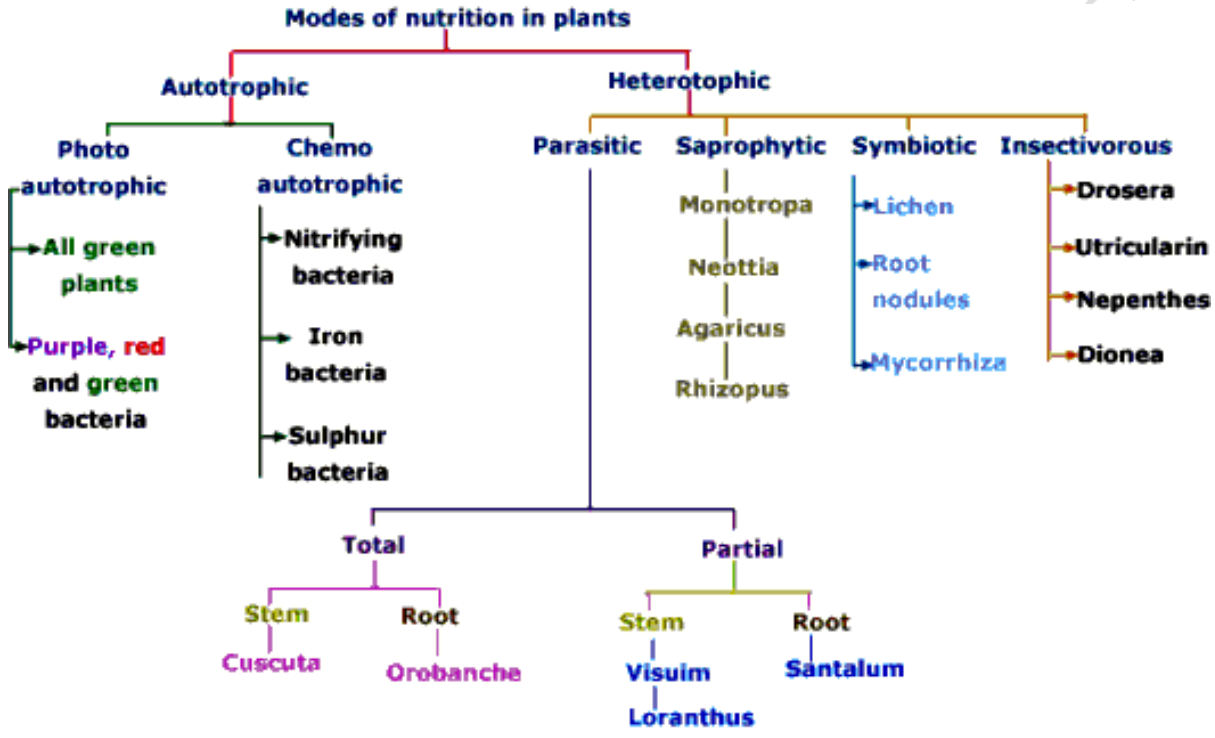
Alkaloids that, like most other nitrogen-containing compounds, are synthesized from amino acids. Alkaloids are especially interesting because they are toxic to both herbivores and humans; yet they have some very important medicinal properties for humans. The nitrogen atom, which in these substances is almost always part of a heterocyclic ring origin, is found innately in the structure of the amino acids from which they came or are the result of the circularization of the given amino acid. This is the case with aspartic acid that combines with glyceraldehyde-3-phosphate in the production of nicotinic acid (a precursor of the alkaloid, nicotine) in plants such as tobacco (Nicotianatabacum). Nicotine is well known as a toxic component of tobacco smoke. There are many categories of alkaloids, including pyrrolidine, tropane, piperidine, pyrrolizidine, quinolizidine, isoquinoline, andindole alkaloids (Michael, 2003, 2004). Much of the carbon skeleton of some of these alkaloids is derived from the mevalonic acid pathway, but it is beyond the scope of this chapter to go into the details of the biosynthesis of all types of alkaloids.

Alkaloids are not solely defensive substances. Some red and yellow alkaloids, called **betalains**, like carotenoids and anthocyanins, act as attractants in flowers and fruits of cacti (e.g., prickly pear cactus,Opuntia spp.). It is interesting to note that plant families that contain betalain pigments (e.g., in the-goose-foot family, Chenopodiaceae, with a familiar example being the root of the garden beet, Betavulgaris) never contain anthocyanins (see Robinson, 1991). Some pyrrolizidine alkaloids act as attractants by mimicking such compounds as the sexual **pheromones** normally produced by some insects like butterflies. These compounds trick the insect into visiting the flower and spreading the plant's pollen, but alkaloids, in general, are toxic. When taken in sufficient quantity, alkaloids are dangerously toxic to humans, but at lower doses, many are helpful — **morphine**, **codeine**, **atropine**, and **ephedrine**, to name a few. Other alkaloids, including **nicotine**, **caffeine**, and **cocaine**, are popular non-medicinal stimulants or sedatives, but they too have their toxic effects

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AUTOTROPHIC AND HETEROTROPHIC GROWTH

The chemical substances that provide nourishment to living organisms are called nutrients. Depending on the mode of nutrition the organisms are classified as autotrophs and heterotrophs.



AUTOTROPHIC NUTRITION

Autotrophs synthesize organic materials from inorganic materials. Some organisms derive their energy for this process from sunlight and are called photoautotrophs, e.g. Green plants.

Other organisms use chemical energy and are called chemoautotrophs, e.g. Nitrifying bacteria. Photoautotrophs are the primary producers in food chains.

HETEROTROPHIC NUTRITION

Heterotrophs eat readymade complex organic food. From this they obtain energy for metabolism, atoms and molecules to build new protoplasm or repair worn out parts, and ions, co-enzymes and vitamins vital for chemical processes. There are four types of heterotrophic nutrition - saprophytic, parasitic, symbiotic and insectivorous.

PARASITIC PLANTS

The non-green plants which live on other living organisms and obtain their food from them are called **parasitic plants**. A common plant parasite is Cuscuta or dodder plant (amarbel). This parasite produces yellow tubular structures which attach themselves to a plant and wrap them around the stem and branches. Its leaves are reduced to minute scales. Chlorophyll is not present in this plant. It absorbs readymade food from the host plant on which it climbs, through special structures called **haustoria**. The haustoria absorb food from the host plant.

- **Total stem parasite:** Cuscata is a total stem parasite. They absorb food material from host by parasitic root called haustoria.
- **Partial stem parasite** - Loranthus, Viscum
- Balanophora is **total root parasite**.
- **Partial root parasite** is Santalum, sandal wood tree, an evergreen plant which grows at many places in India. Roots develop haustoria that make contact with roots of Dalbergia, Eucalyptus etc.

INSECTIVOROUS PLANTS

They are said to be partly autotrophic and partly heterotrophic. These autotrophs supplement their nutritional requirements by trapping and digesting insects and other small animals. The trapped insects are killed and their proteins are digested by proteolytic enzymes secreted by the epidermis of the leaf. They can live without insects but their growth is stimulated when they digest the insects. The leaves of these plants are modified in several ways for trapping insects, e.g. Pitcher plants, Drosera, Utricularia and Dionea

SYMBIOTIC PLANTS

Plants which live in association with other plants and share shelter and nutrients are called **symbiotic plants**. This association between two different plants is called **symbiotic association** or **relationship**. Both the plants benefit from a symbiotic association.

Examples:

- Certain fungi live in the roots of trees. The tree provides nutrients to the fungus. In return, the fungus provides certain nutrients from the soil to the tree.
- In lichens also, symbiotic relationship is seen. There is an association between a green alga and a non-green fungus. The fungus provides shelter, water and minerals to the alga. The alga, in return, provides food to the fungus. The alga, being green, prepares food by photosynthesis.

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PLANT GROWTH REGULATORS

Traditionally, five major hormone classes are described: auxins, cytokinins, gibberellins, abscisic acid and ethylene. Other active compounds have been known for years, and there are new-comers with increasingly strong claims for inclusion. These include brassinosteroids and jasmonic acid in particular, but also polyamines, salicylic acid and oligosaccharides; for details of these, some informative review articles are listed at the end of the chapter.

Most hormones have certain biochemical trends in common: small molecules synthesised from ubiquitous precursors (amino acids, mevalonic acid, nucleotides) sometimes via multi-step pathways, then deactivated by oxidation or conjugation (linking to other small molecules such as glucose and amino acids).

AUXINS

Auxin in its most common natural form indole-3-acetic acid (IAA), was the first plant hormone to be isolated, and was long thought to be derived exclusively from the amino acid tryptophan. Plants and certain plant pathogenic bacteria synthesise IAA, although the genes, enzymes and reaction intermediates differ between prokaryote and eukaryote. Active growing tissues, especially shoot tips and young leaves, synthesise auxins, as do developing fruits and seeds. Roots appear to produce much less auxin, but auxin has vital functions in lateral root development. Plants and bacteria can deactivate auxins by irreversible oxidation involving enzymes such as IAA oxidase, or by covalently linking (conjugating) them to other small molecules: sugars, cyclitols, amino acids. Some conjugates (e.g. IAA-aspartate) act as inactive auxin stores, regenerating active auxin when the link is hydrolysed.

Auxins promote both cell division and cell growth. The most important naturally occurring auxin is indole-3-acetic acid (IAA), but its use in plant cell culture media is limited because it is unstable to both heat and light. It is more common, though, to use stable chemical analogues of IAA as a source of auxin in plant cell culture media. 2,4-Dichlorophenoxyacetic acid (2,4-D) is the most commonly used auxin and is extremely effective in most circumstances.

Effects

Auxin participates in phototropism, geotropism, hydrotropism and other developmental changes. The uneven distribution of auxin, due to environmental cues, such as unidirectional light or gravity force, results in uneven plant tissue growth, and generally, auxin governs the form and shape of plant body, direction and strength of growth of all organs, and their mutual interaction.
Auxin stimulates cell elongation by stimulating wall-loosening factors, such as elastins, to loosen cell walls. The effect is stronger if gibberellins are also present. Auxin also stimulates cell division if cytokinins are present. When auxin and cytokinin are applied to callus, rooting can be generated if the auxin concentration is higher than cytokinin concentration. Xylem tissues can be generated when the auxin concentration is equal to the cytokinins.

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Auxin also induces sugar and mineral accumulation at the site of application.

Root Growth And Development

Auxins promote root initiation.Auxin induces both growth of pre-existing roots and adventitious root formation, i.e., branching of the roots. As more native auxin is transported down the stem to the roots, the overall development of the roots is stimulated. If the source of auxin is removed, for example the tips of stems are trimmed, the roots are less stimulated accordingly, and growth of stem is supported instead.

In horticulture, auxins, especially NAA and IBA, are commonly applied to stimulate root initiation when rooting cuttings of plants. However, high concentrations of auxin inhibit root elongation and instead enhance adventitious root formation. Removal of the root tip can lead to inhibition of secondary root formation.

Apical Dominance

Auxin induces shoot apical dominance; the axillary buds are inhibited by auxin, as a high concentration of auxin directly stimulates ethylene synthesis in lateral buds, causing inhibition of their growth and potentiation of apical dominance. When the apex of the plant is removed, the inhibitory effect is removed and the growth of lateral buds is enhanced.

Auxin Effects In Tissue Culture

In tissue culture, depending on other hormones present in the medium, changes in auxin concentrations may change the type of growth, e.g., stimulation of root formation may switch to callus induction etc. In this respect, each tissue culture system is unique, and the effects of different concentrations of auxins and other hormones must be tested for each case individually and only to some extent can the results can be transferred to other cultures.

Induction Of Callus Growth

An auxin is generally required for the induction of callus from explants. For callus induction from broadleaved trees, 2,4-D is generally used.

Auxins promote cell dispersion in suspension cultures while cytokinins tend to cause cell aggregation.

The relatively high levels of auxin added to liquid media to obtain dispersion will prevent morphogenesis, but might induce embryogenesis if the cells are still competent. Whereas Cytokinins tend to promote the formation of chlorophyll in callus and suspension cultures, auxins can be inhibitory.

Organ Cultures

An auxin is almost invariably required to promote the initial growth of meristem and shoot tip explants.

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Embryogenesis

The process of somatic embryogenesis is often initiated in media containing high levels of auxins(especially 2,4-D), but embryos usually do not develop further until the auxin concentration is reduced.

CYTOKININS

In many ways, cytokinins are opposites of auxins, being synthesised in roots but with most dramatic effects on shoot development. However, shoot tissues can also produce cytokinins, as can developing seeds. A classic example of the latter is coconut milk, the copious liquid endosperm from coconut seed, which is still a popular cytokinin source in plant tissue culture media. Cytokinins were originally named from their ability to promote cell division, but they also function in initiation of new shoot structures, dormancy release and retardation of senescence.

Cytokinins are derivatives of adenine, one of the purine bases found in all DNA and RNA. Instead, a de novo pathway using free adenine nucleotides as substrate appears to predominate.

Cytokinins promote cell division. Naturally occurring cytokinins are a large group of structurally related purine derivatives. Of the naturally occurring cytokinins, two have some use in plant tissue culture media, zeatin and N6-(2-isopentyl) adenine (2iP). Their use is not widespread as they are expensive (particularly zeatin) and relatively unstable. The synthetic analogues kinetin and 6-benzylaminopurine (BAP) are therefore used more frequently. Non-purine-based chemicals, such as substituted phenylureas, are also used as cytokinins in plant cell culture media. These substituted phenylureas can also substitute for auxin in some culture systems.

Commonly used cytokinins, their abbreviations and chemical names:

Abbreviation/name	Chemical name
BAP[a]	6-Benzylaminopurine
2iP (or IPA)[b]	N^6-(2-Isopentyl)adenine
Kinetin[a]	6-Furfurylaminopurine
Thidiazuron[c]	1-Phenyl-3-(1,2,3-thiadiazol-5-yl)urea
Zeatin[b]	4-Hydroxy-3-methyl-*trans*-2-butenylaminopurine

[a]Synthetic analogues.
[b]Naturally occurring cytokinins.
[c]A substituted phenylurea-type cytokinin.

Cytokinins Action

Cytokinins promote cell division in plants. Produced in the developing shoots, roots, fruits and seeds of a plant, cytokinins are very important in the culturing of plant tissues in the laboratory. A high ratio of auxins to cytokinins in a tissue-culture medium stimulates root formation. A low

ratio promotes shoot formation. Cytokinins are also used to promote lateral bud growth in flowering plants.

Adventitious Shoot Formation

Cytokinins are very effective in promoting director indirect shoot initiation.

Embryogenesis

A low concentration of cytokinin (typically 0.5-2.5 μM) is often added to media for the induction of embryogenic callus, especially in broad-leafed plants.

Adventitious Shoot Bud Formation

The formation of adventitious shoots, whether directly from explanted tissues, or indirectly from callus, is regulated by an interaction between auxins and cytokinins.

Inhibition Of Root Formation

High concentrations of cytokinin (0.5-10 mg/l) generally inhibit or delay root formation and also prevent root growth and the promotive effects of auxins on root initiation.

GIBBERELLINS

Gibberellins were first noticed through symptoms of a disease (known in Japanese as bakanae = foolish seedling) on rice that caused excessive stem elongation. The causative agent, a fungus called Gibberella fujikuroi, contains several different types of gibberellin (abbreviated to GA, after gibberellic acid, the first form discovered), some of which also exist in plants. Plants possess many other unique gibberellins, and collectively there are well over 100 identified compounds. This intimidating complexity can be reduced to a comprehensible level by realising that each species contains only about 25 of these gibberellins and that most gibberellins are biosynthetic intermediates or inactive end-products, and not active in their own right. There are many steps and enzymes involved in building up the 19- and 20-carbon gibberellin molecules from five-carbon mevalonic acid. Several parallel pathways exist differing only in number of hydroxyl (–OH) groups. Hydroxyl groups are the key to gibberellin functions: some positions (3b) are generally essential for activity, whereas others (2b) completely abolish it. Inactivation by conjugation to glucose also occurs. Gibberellin synthesis takes place mainly in developing leaves and stems, in developing seeds and during germination. Gibberellins function in dormancy release and germination, as well as in growth promotion (e.g. stem elongation, fruit tissue expansion).

They are involved in regulating cell elongation, and are agronomically important in determining plant height and fruit-set. Only a few of the gibberellins are used in plant tissue culture media, GA3 being the most common.

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ABSCISIC ACID

Abscisic acid (ABA) is an unfortunate name because this hormone has little to do with abscission. But once again it tells a story: cotton, one of the plants originally studied, turns out to be an exception in that ABA does promote fruit shedding. ABA is a 15-carbon molecule and its synthesis occurs from breakdown of carotenoid pigments, especially violaxanthin, a 40-carbon molecule. Previously, mevalonic acid was thought to be the main precursor, with early steps in common with gibberellin biosynthesis. This alternative pathway may operate in tissues such as avocado mesocarp and in tomato seedlings. ABA is produced in large quantities in water-stressed tissues, especially roots and leaves, but also has a role in seed maturation, dormancy and senescence. ABA concentrations are lowered by oxidative deactivation to phaseic acid or by formation of glucosides.

Abscisic acid (ABA) inhibits cell division. It is most commonly used in plant tissue culture to promote distinct developmental pathways such as somatic embryogenesis.

ETHYLENE

Ethylene (= ethene; C_2H_4) is a unique gaseous hormone that diffuses rapidly out of plant tissues. Its immediate precursor is 1-aminocyclopropane-1-carboxylate (ACC) which in turn originates from S-adenosyl methionine, a derivative of another common amino acid:

Methionine **Enzymes:**

↓ *SAM synthase*

S-adenosyl-methionine

↓ *ACC synthase*

ACC

↓ *ACC oxidase*

Ethylene

Ethylene is produced in response to cell damage and other stresses such as anoxia. It accumulates rapidly during fruit ripening and senescence, but all living cells produce some ethylene. Oxidation and conjugation can occur, but dissipation into the atmosphere is probably the main 'means of disposal'.

Ethylene is a gaseous, naturally occurring, plant growth regulator most commonly associated with controlling fruit ripening in climacteric fruits, and its use in plant tissue culture is not widespread. It does, though, present a particular problem for plant tissue culture. Some plant cell

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cultures produce ethylene, which, if it builds up sufficiently, can inhibit the growth and development of the culture. The type of culture vessel used and its means of closure affect the gaseous exchange between the culture vessel and the outside atmosphere and thus the levels of ethylene present in the culture.

NOTE:

Auxins and cytokinins are the most widely used plant growth regulators in plant tissue culture and are usually used together, the ratio of the auxin to the cytokinin determining the type of culture established or regenerated (see Figure 2.1). A high auxin to cytokinin ratio generally favours root formation, whereas a high cytokinin to auxin ratio favours shoot formation. An intermediate ratio favours callus production.

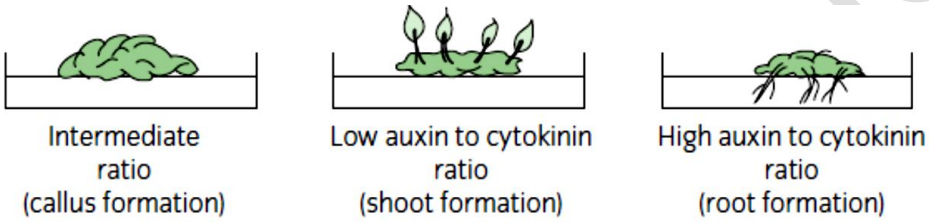
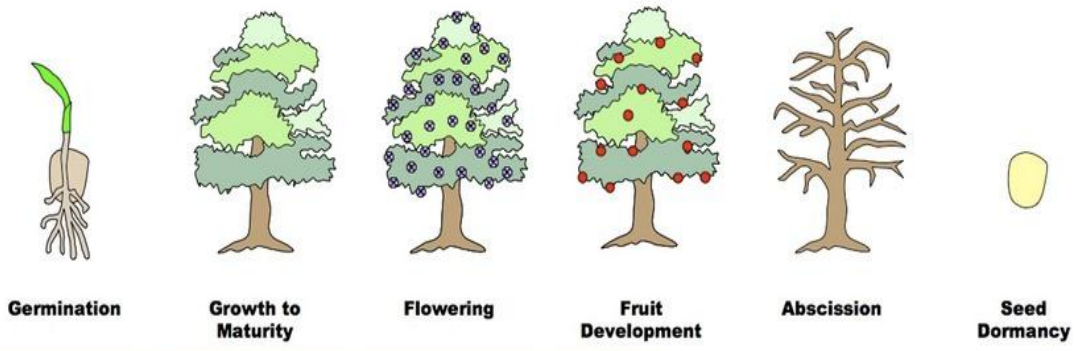


Figure 2.1 The effect of different ratios of auxin to cytokinin on the growth and morphogenesis of callus. High auxin to cytokinin ratios promote root development, while low ratios promote shoot development. Intermediate ratios promote continued growth of the callus without differentiation.



Gibberellin							
Auxin							
Cytokinins							
Ethylene							
ABA							

CELL SUSPENSION CULTURE

Cultures are generally initiated from sterile pieces of a whole plant. These pieces are termed **explants**, and may consist of pieces of organs, such as leaves or roots, or may be specific cell types, such as pollen or endosperm. Many features of the explant are known to affect the efficiency of culture initiation. Generally, younger, more rapidly growing tissue (or tissue at an early stage of development) is most effective. Several different culture types most commonly used in plant transformation studies will now be examined in more detail.

CALLUS

Explants, when cultured on the appropriate medium, usually with both an auxin and a cytokinin, can give rise to an unorganized, growing, and dividing mass of cells. It is thought that any plant tissue can be used as an explant, if the correct conditions are found. In culture, this proliferation can be maintained more or less indefinitely, provided that the callus is subcultured on to fresh medium periodically. During callus formation, there is some degree of dedifferentiation (i.e. the changes that occur during development and specialization are, to some extent, reversed), both in morphology (a callus is usually composed of unspecialized parenchyma cells) and metabolism. One major consequence of this dedifferentiation is that most plant cultures lose the ability to photosynthesize. This has important consequences for the culture of callus tissue, as the metabolic profile will probably not match that of the donor plant. This necessitates the addition of other components — such as vitamins and, most importantly, a carbon source—to the culture medium, in addition to the usual mineral nutrients.

Callus culture is often performed in the dark (the lack of photosynthetic capability being no drawback) as light can encourage differentiation of the callus. During long term culture, the culture may lose the requirement for auxin and/or cytokinin. This process, known as **habituation**, is common in callus cultures from some plant species (such as sugar beet).

Callus cultures are extremely important in plant biotechnology. Manipulation of the auxin to cytokinin ratio in the medium can lead to the development of shoots, roots, or somatic embryos from which whole plants can subsequently be produced. Callus cultures can also be used to initiate cell suspensions, which are used in a variety of ways in plant transformation studies.

CELL-SUSPENSION CULTURES

Callus cultures, broadly speaking, fall into one of two categories: compact or friable. In compact callus, the cells are densely aggregated, whereas in friable callus, the cells are only loosely associated with each other and the callus becomes soft and breaks apart easily. Friable callus provides the inoculum to form cell-suspension cultures. Explants from some plant species or particular cell types tend not to form friable callus, making it difficult to initiate cell suspension. The friability of the callus can sometimes be improved by manipulating the medium components

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or by repeated subculturing. The friability of the callus can also sometimes be improved by culturing it on semi-solid medium (medium with a low concentration of gelling agent).

When friable callus is placed into a liquid medium (usually the same composition as the solid medium used for the callus culture) and then agitated, single cells and/or small clumps of cells are released into the medium. Under the correct conditions, these released cells continue to grow and divide, eventually producing a cell-suspension culture. A relatively large inoculum should be used when initiating cell suspensions so that the released cell numbers build up quickly. The inoculum should not be too large though, as toxic products released from damaged or stressed cells can build up to lethal levels. Large cell clumps can be removed during subculture of the cell suspension.

Cell suspensions can be maintained relatively simply as batch cultures in conical flasks. They are continually cultured by repeated subculturing into fresh medium. This results in dilution of the suspension and the initiation of another batch growth cycle. The degree of dilution during subculture should be determined empirically for each culture. Too great a degree of dilution will result in a greatly extended lag period or, in extreme cases, death of the transferred cells. After subculture, the cells divide and the biomass of the culture increases in a characteristic fashion, until nutrients in the medium are exhausted and/or toxic byproducts build up to inhibitory levels: this is called the **stationary phase**. If cells are left in the stationary phase for too long, they will die and the culture will be lost. Therefore, cells should be transferred as they enter the stationary phase. It is therefore important that the batch growth-cycle parameters are determined for each cell suspension culture

BATCH CULTURES

Batch cultures are initiated by inoculating cells into a fixed volume of nutrient medium. As growth proceeds, the amounts of cell material increases until nutrients in the medium are depleted or there is the accumulation of an inhibitory metabolite. Batch cultures have a number of disadvantages that restrict their suitability for extended studies of growth and metabolism, or for the industrial production of plant cells, but they are nevertheless widely used for many laboratory investigations. Small cultures are frequently agitated on orbital shakers onto which are fixed suitable containers, which range in volume from 100 ml(Erlenmeyer conical flasks) to 1000 ml (spherical flasks); the quantity of medium being approximately the same as the flask volume. The shakers are usually operated at speeds from 30-180 rpm with an orbital motion of about 3 cm. Alternatively, stirred systems can be used.

CONTINUOUS CULTURES

Using batch cultures, it is difficult to obtain a steady rate of production of new cells having constant size and composition. Attempts to do so necessitate frequent sub-culturing, at intervals equivalent to the doubling time of the cell population. Satisfactorily balanced growth can only be produced in continuous culture, a method, which is especially important when plant cells are to be used for the large-scale production of a primary or secondary metabolite. Continuous culture techniques require fairly complicated apparatus. Agitation of larger cultures in bio-reactors is

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usually achieved by stirring with a turbine and/or by passing sterile air (or a controlled gaseous mixture) into the culture from below and releasing it through plugged vents. Mechanically stirred reactors damage plant cells by shearing. This is minimised in air-lift reactors.

THE USE OF SUSPENSION CULTURES IN PLANT PROPAGATION

The growth of plant cells is more rapid in suspension than in callus culture and is also more readily controlled because the culture medium can be easily amended or changed. Organs can be induced to develop in cell suspensions: root and shoot initiation usually commences in cell aggregates. Somatic embryos may arise from single cells. Cells from suspensions can also be plated onto solid media where single cells and/or cell aggregates grow in to callus colonies from which plants can often be regenerated. For these reasons suspension cultures might be expected to provide a means of very rapid plant multiplication. There are two methods:

- Plants may be obtained from somatic embryos formed in suspensions. Once embryos have been produced, they are normally grown into plantlets on solid media, although other methods are potentially available;
- Cells from suspensions are plated onto solid media where single cells and/or cell aggregates grow into callus colonies from which plants can often be regenerated.

PROTOPLASTS

Protoplasts are plant cells with the cell wall removed. Protoplasts are most commonly isolated from either leaf mesophyll cells or cell suspensions, although other sources can be used to advantage. Two general approaches to removing the cell wall (a difficult task without damaging the protoplast) can be taken: mechanical or enzymatic isolation. Mechanical isolation, although possible, often results in low yields, poor quality, and poor performance in culture due to substances released from damaged cells. Enzymatic isolation is usually carried out in a simple salt solution with a high osmoticum, plus the cell-wall-degrading enzymes. It is usual to use a mix of both cellulase and pectinase enzymes, which must be of high quality and purity.

Protoplasts are fragile and damaged easily, and therefore must be cultured carefully. Liquid medium is not agitated and a high osmotic potential is maintained, at least in the initial stages. The liquid medium must be shallow enough to allow aeration in the absence of agitation. Protoplasts can be plated out on to solid medium and callus produced. Whole plants can be regenerated by organogenesis or somatic embryogenesis from this callus. Protoplasts are ideal targets for transformation by a variety of means.

ROOT CULTURES

Root cultures can be established in vitro from explants of the root tip of either primary or lateral roots and can be cultured on fairly simple media. The growth of roots in vitro is potentially unlimited, as roots are indeterminate organs. Although the establishment of root cultures was one of the first achievements of modern plant tissue culture, they are not widely used in plant transformation studies.

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SHOOT TIP AND MERISTEM CULTURE

The tips of shoots (which contain the shoot apical meristem) can be cultured in vitro, producing clumps of shoots from either axillary or adventitious buds. This method can be used for clonal propagation. Shoot meristem cultures are potential alternatives to the more commonly used methods for cereal regeneration as they are less genotype dependent and more efficient (seedlings can be used as donor material).

EMBRYO CULTURE

Embryos can be used as explants to generate callus cultures or somatic embryos. Both immature and mature embryos can be used as explants. Immature, embryo-derived embryogenic callus is the most popular method of monocotyledon plant regeneration.

MICROSPORE CULTURE

Haploid tissue can be cultured in vitro by using pollen or anthers as an explant. Pollen contains the male gametophyte, which is termed the **microspore**. Both callus and embryos can be produced from pollen. Two main approaches can be taken to produce cultures in vitro from haploid tissue.

The first method depends on using the anther as the explant. Anthers (somatic tissue that surrounds and contains the pollen) can be cultured on solid medium (agar should not be used to solidify the medium as it contains inhibitory substances). Pollen-derived embryos are subsequently produced via dehiscence of the mature anthers. The dehiscence of the anther depends both on its isolation at the correct stage and on the correct culture conditions. In some species, the reliance on natural dehiscence can be circumvented by cutting the wall of the anther, although this does, of course, take a considerable amount of time. In the second method, anthers can also be cultured in liquid medium, and pollen released from the anthers can be induced to form embryos, although the efficiency of plant regeneration is often very low. Immature pollen can also be extracted from developing anthers and cultured directly, although this is a very time-consuming process.

Both methods have advantages and disadvantages. Some beneficial effects to the culture are observed when anthers are used as the explant material. There is, however, the danger that some of the embryos produced from another culture will originate from the somatic anther tissue rather than the haploid microspore cells. If isolated pollen is used there is no danger of mixed-embryo formation, but the efficiency is low and the process is time-consuming. In microspore culture, the condition of the donor plant is of critical importance, as is the timing of isolation. Pretreatments, such as a cold treatment, are often found to increase the efficiency. These pretreatments can be applied before culture, or, in some species, after placing the anthers in culture.

Plant species can be divided into two groups, depending on whether they require the addition of plant growth regulators to the medium for pollen/anther culture; those that do also often require

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organic supplements, such as amino acids. Many of the cereals (rice, wheat, barley, and maize) require medium supplemented with plant growth regulators for pollen or another culture.

Regeneration from microspore explants can be obtained by direct embryogenesis, or via a callus stage and subsequent embryogenesis. Haploid tissue cultures can also be initiated from the female gametophyte (the ovule). In some cases, this is a more efficient method than using pollen or anthers. Plants obtained from haploid cultures may not be haploid. This can be a consequence of chromosome doubling during the culture period. Chromosome doubling (which often has to be induced by treatment with chemicals such as colchicine) may be an advantage, as in many cases haploid plants are not the desired outcome of regeneration from haploid tissues. Such plants are often referred to as **di-haploids**, because they contain two copies of the same haploid genome.

PATTERNS OF GROWTH AND DIFFERENTIATION

A typical unorganised plant callus, initiated from a new explant or a piece of a previously-established culture, has three stages of development, namely:

- The induction of cell division;
- A period of active cell division during which differentiated cells lose any specialised features they may have acquired and become dedifferentiated;
- A period when cell division slows down or ceases and when, within the callus, there is increasing cellular differentiation.

These phases are similarly reproduced by cell suspensions grown in a finite volume of medium (a batch culture), where according to a variety of different parameters that can be used to measure growth (e.g. cell number, cell dry weight, total DNA content) an S-shaped growth curve is generally obtained. The phases are:

- A lag phase;
- A period of exponential and then linear growth;
- A period when the rate of growth declines;
- A stationary phase when growth comes to a halt.

Some differentiation of cells may occur in cell cultures during the period of slowed and stationary growth, but generally it is less marked and less complete than that which occurs in callus cultures. Cultures cannot be maintained in stationary phase for long periods. Cells begin to die and, as their contents enter the nutrient medium, death of the whole culture accelerates. Somewhat similar patterns of growth also occur in cultures of organised structures. These also cease growth and become moribund as the components of the medium become exhausted.

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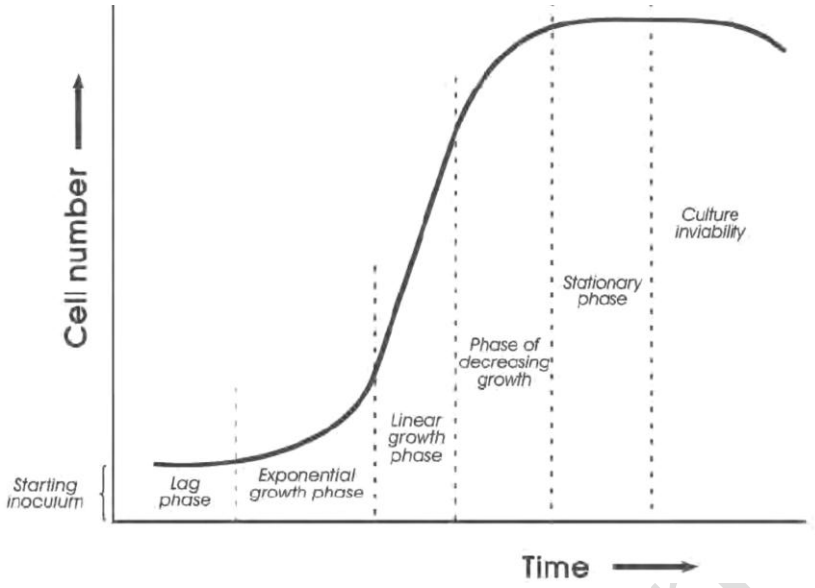


Fig : Diagram showing the phases of growth in batch suspension culture

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PRODUCTION OF SECONDARY METABOLITES BY PLANT SUSPENSION CULTURES

ACCUMULATION OF SECONDARY METABOLITES IN PLANT CELL CULTURES

For plant cell culture techniques to become economically viable, it is important to develop methods that would allow for consistent generation of high yields of products from cultured cells [Berlin and Sasse, 1985]. Careful selection of productive cells and cultural conditions resulted in accumulation of several products in higher levels in cultured cells. In order to obtain yields in high concentrations for commercial exploitation, efforts have focused on the stimulation of biosynthetic activities of cultured cells using various methods. Productivity is critical to the practical application of plant cell culture technology to production of plant-specific bioactive metabolites. Until now, various strategies have been developed to improve the production of secondary metabolites using plant cell cultures.

The tissue culture cells typically accumulate large amounts of secondary compounds only under specific conditions. That means maximization of the production and accumulation of secondary metabolites by plant tissue cultured cells requires (i) manipulating the parameters of the environment and medium, (ii) selecting high yielding cell clones, (iii) precursor feeding, and (iv) elicitation.

- **Optimization of cultural conditions**: Number of chemical and physical factors like media components, phytohormones, pH, temperature, aeration, agitation, light affecting production of secondary metabolites has been extensively studied.
- **Selection of high-producing strains: P**lant cell cultures represent a heterogeneous population in which physiological characteristics of individual plant cells are different. Synthesis of several products in high amounts using selection and screening of plant cell cultures.
- **Precursor feeding**: Exogenous supplies of a biosynthetic precursor to culture medium mayalso increase the yield of the desired product.
- **Elicitation**: plants produce secondary metabolites in nature as a defense mechanism against attack by pathogens. Elicitors are signals triggering the formation of secondary metabolites. Use of elicitors of plant defense mechanisms, i.e. elicitation, has been one of the most effective strategies for improving the productivity of bioactive secondary metabolites.

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HAIRY ROOT CULTURES AND THEIR CULTIVATION

Hairy root is a plant disease caused by Agrobacterium rhizogenes, a Gram-negative soil bacterium. When the bacterium infects the plant, the T-DNA between the TR and TL regions of the Ri-plasmid in the bacterium is transferred and integrated into the nuclear genome of the host plant. The transformation process produces a valuable by-product, hairy root, which will form at or near the site of infection. In addition, opines are produced and serve as specific food for the bacteria. Hairy roots grow rapidly, show plagiotropic growth, and are highly branched on phytohorm one-free medium. The transformed root is highly differentiated and can cause stable and extensive production of secondary metabolites, whereas other plant cell cultures have a strong tendency to be genetically and biochemically unstable and often synthesize very low levels of useful secondary metabolites. Most importantly, A. rhizogenescan transfer T-DNA from binary vectors and enable the production of transgenic plants containing foreign genes carried on a second plasmid. This property has been used to produce transgenic plants.

HAIRY ROOT INDUCTION AND SELECTION

ESTABLISHMENT OF A HAIRY ROOT CULTURE SYSTEM

To succeed in establishing a hairy root culture system for a certain plant species, several essential conditions should be taken into consideration. These conditions include the bacterial strain of A. rhizogenes, an appropriate explants, a proper antibiotic to eliminate redundant bacteria after cocultivation, and a suitable culture medium. Based on the types of opines produced, the strains of A.rhizogenes can be separated into five lines: octopine, agropine, nopaline, mannopine, and cucumopine (Zhou et al. 1998). Agropinestrains are the most often used strains owing to their strongest induction ability. Most plant materials, such as hypocotyls, leaf, stem, stalk, petiole, shoot tip, cotyledon, protoplast, storage root,or tuber, can be used to induce hairy roots (Mugnier 1988; Han etal. 1993; Drewes et al. 1995; Giri et al. 2001; Krolicka et al. 2001;Azlan et al. 2002). However, for different species, the proper ex plant material may vary and the age of the material is most critical, with juvenile material being optimal. To induce hairy root, explants are separately wounded and co cultivated or inoculated with A. rhizogenes. Usually two or three days later, the explants scan be transferred into solid media with antibiotics, such ascefotaxime sodium, carbencilin disodium, vancomycin, ampicillinsodium, claforan, streptomycin sulphate, or tetracycline, ranging in concentration from 100 to 500 g/mL, to kill or eliminate redundant bacteria (Spano et al. 1981; Drewes et al. 1995; Giri et al.2001; Krolicka et al. 2001). The hairy roots will be induced within a short period of time, which varies from one week to over a month depending on different plant species. The decontaminated hairy roots can be sub cultured on phytohormone-free medium. Optimizing the composition of nutrients for hairy root cultures is critical to gain a high production of secondary metabolites. Factors such as the carbon source and its concentration, the ionic concentration of the medium, the pH of the medium, light,

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phytohormones, temperature, and inoculums are known to influence growth and secondary metabolism (Christen et al. 1992; Toivonen et al. 1992; Rhodes et al. 1994; Arroo et al. 1995; Bhadraet al. 1995; Vanhala et al. 1998; Morgan et al. 2000). Heavy metal ions and the concentrations of phosphate, nitrate, and ammonia have also been well studied (Payne et al. 1987; Toivonen et al.1991; Christen et al. 1992; Sevon et al. 1992). The addition ofauxin and elicitors often increases the levels of secondary metabolites (Dymov et al. 1997; Pittaalvarez et al. 1998; Rijhwani et al. 1998; Singh et al. 1998; Vanhala et al. 1998). Because of thesefactors and the fact that individual hairy roots may have different requirements for nutrient conditions, the culture conditions should be optimized separately for each species and for individual clones.

REPORTER GENE

The glucuronidase (GUS) gene is usually transferred into hairy roots as a reporter gene and it can be analyzed easily by histological assay. So far, the GUS reporter system is the most common means of monitoring plant systems. In other cases, neomycin phosphotransferase II(NPT-II) encoding the kanamycin-resistance enzyme has been used. Sometimes, both GUS and NPTII have been transferred into the hairy roots. Recently, the gene for green fluorescent protein (GFP) was used successfully as a reporter gene in Catharanthus roseus L. hairy roots.

SELECTION OF HAIRY ROOT

Owing to the site uncertainty of T-DNA integration into the host plant genome, the hairy roots derived often show different accumulation patterns of secondary metabolites. Mano et al. (1989) analyzed 45 hairy root clones of Duboisia leichhardtii and found that there was considerable variation in growth rate, alkaloid content, and productivity among the clones. Generally, hairy roots are considered to be stable and easy to subculture. Nonetheless, hairy roots also possess a certain amount of heterogeneity, even if derived from a single root tip, and the repeated selection seems to be important to obtain high scopolamine-producing hairy root lines.

APPLICATION OF HAIRY ROOTS

- Functional analysis of genes
- Expressing foreign proteins
- Production of secondary metabolites: Normally, root cultures need an exogenous phytohormone supply and grow very slowly, resulting in the poor or negligible synthesis of secondary metabolites. The hairy root system is stable and highly productive under hormone-free culture conditions. The fast growth, low doubling time, ease of maintenance, and ability to synthesize a range of chemical compounds of hairy root cultures offer additional advantages as continuous sources for the production of valuable secondary metabolites. Hairy roots are also valuable sources of photochemical that are useful as pharmaceuticals, cosmetics, and food additives.

POTENTIAL PROBLEMS

- Different regulation of secondary metabolism in related species.

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- Over expression of key enzymes does not always improve secondary metabolism
- Possible reduction of chromosome numbers during subculture
- Co suppression of endogenous and foreign genes

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TECHNIQUES IN RAISING TRANSGENCIES

WHAT ARE TRANSGENIC PLANTS?

A transgenic crop plant contains a gene or genes which have been artificially inserted instead of the plant acquiring them through pollination. The inserted gene sequence (known as the **transgene**) may come from another unrelated plant, or from a completely different species: transgenic Bt corn, for example, which produces its own insecticide, contains a gene from a bacterium. Plants containing transgenes are often called **genetically modified** or **GM crops**, although in reality all crops have been genetically modified from their original wild state by domestication, selection and controlled breeding over long periods of time. On this web site we will use the term **transgenic** to describe a crop plant which has transgenes inserted.

WHY MAKE TRANSGENIC CROP PLANTS?

A plant breeder tries to assemble a combination of genes in a crop plant which will make it as useful and productive as possible. Depending on where and for what purpose the plant is grown, desirable genes may provide features such as higher yield or improved quality, pest or disease resistance, or tolerance to heat, cold and drought. Combining the best genes in one plant is a long and difficult process, especially as traditional plant breeding has been limited to artificially crossing plants within the same species or with closely related species to bring different genes together. For example, a gene for protein in soybean could not be transferred to a completely different crop such as corn using traditional techniques. Transgenic technology enables plant breeders to bring together in one plant useful genes from a wide range of living sources, not just from within the crop species or from closely related plants. This technology provides the means for identifying and isolating genes controlling specific characteristics in one kind of organism, and for moving copies of those genes into another quite different organism, which will then also have those characteristics. This powerful tool enables plant breeders to do what they have always done - generate more useful and productive crop varieties containing new combinations of genes - but it expands the possibilities beyond the limitations imposed by traditional cross-pollination and selection techniques.

Agro bacterium mediated gene transfer or vector mediated gene transfer or indirect method of gene transfer.

Agrobacterium mediated gene transfer is one of the most widely used methods of gene transfer. Agrobacterium tumefaciens is a bacterial plant pathogen found in the soil which causes crown gall disease in plants.

The basic mechanism that works behind it is, transfer of a segment or small piece of bacterial DNA to the plant genome. This bacterial DNA segment gets incorporated in the plant genome, stabilizes there and transcribes or expresses itself.

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The bacterial DNA causing disease is situated on a plasmid, Ti plasmid (Tumour inducing plasmid, 200kb) instead of on a bacterial chromosome. The segment which is actually transferred to plant genome is situated on T-DNA (transferred DNA, 23kb) of this plasmid.

In the expression of disease symptom, two major sets of genes are involved

- Genes on T-DNA (total 15 genes)
- Virulence genes, i.e., vir genes

Both of the above sets of gene can be present on the same plasmid (co-integrative vectors) or separate plasmid (binary vector).

GENES ON T-DNA

The T-DNA contains two groups / types of genes, which possess the ability to express in plants.

The two types of gene are:

- **Oncogenes –** Encode for synthesis of auxins and cytokinins (phytohormones). The overproduction of phytohormones leads to proliferation of callus or tumour formation.
- **Opine synthesizing genes –** Encode for synthesis of opines (a product from amino acids + sugars, which are produced and excreted by the crown gall infected cells and consumed by A. tumefaciens as carbon and nitrogen sources this means opines act as source of nutrient for bacteria.

VIRULENCE GENES

- These are located on vir region (30 kb) of Ti plasmid.
- These genes are not transferred to the plant genome, just help in the process; organized in six operons that are essential for the transfer (virA, virB, virD, and virG) or increase transfer efficiency (virC and virE) of T-DNA.
- They are involved in activities like recognition of host plant, the chemo taxis, attachment to the plant cell, transfer of DNA in plant cell nucleus and integration of inserted DNA into the plant chromosome.

PROCEDURE

- For transformation oncogenic T-DNA is removed and replaced by another desired DNA segment with border sequence.
- While working binary vectors the vir genes are removed and the desired DNA is integrated between the borders.
- The vir genes necessary for transfer to the plant genome are arranged on a second plasmid with T-DNA and both borders removed.

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- Such binary vectors restrict the transferred Ti plasmid from spreading uncontrollably as genes responsible for that are do not get transferred.

ADVANTAGES OF AGROBACTERIUM MEDIATED GENE TRANSFER

Some of the merits of the method are listed below:

- Simple and comparatively less expensive
- High transformation efficiency
- Transgenic crop obtained have better fertility percentage
- Today, protocols for both monocotyledons and dicotyledons are available
- Relatively large length chromosomal segment can be transferred with little arrangement

DISADVANTAGES OF AGROBACTERIUM MEDIATED GENE TRANSFER

- Some of the demerits of the method are listed below
- Time consuming
- Not all kind of cells can be treated by this method
- Cannot transform organelles
- Sometimes leads to false positive results

VECTORLESS OR DIRECT GENE TRANSFER

- In the direct gene transfer methods, the foreign gene of interest is delivered into the host plant cell without the help of a vector. The methods used for direct gene transfer in plants are:
  Chemical mediated gene transfer e.g. chemicals like polyethylene glycol (PEG) and dextran sulphate induce DNA uptake into plant protoplasts. Calcium phosphate is also used to transfer DNA into cultured cells.
- *Microinjection* where the DNA is directly injected into plant protoplasts or cells (specifically into the nucleus or cytoplasm) using fine tipped (0.5 - 1.0 micrometer diameter) glass needle or micropipette. This method of gene transfer is used to introduce DNA into large cells such as oocytes, eggs, and the cells of early embryo.
- *Electroporation* involves a pulse of high voltage applied to protoplasts/cells/ tissues to make transient (temporary) pores in the plasma membrane which facilitates the uptake of foreign DNA.

The cells are placed in a solution containing DNA and subjected to electrical shocks to cause holes in the membranes. The foreign DNA fragments enter through the holes into the cytoplasm and then to nucleus.

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Particle gun/Particle bombardment - In this method, the foreign DNA containing the genes to be transferred is coated onto the surface of minute gold or tungsten particles (1-3 micrometers) and bombarded onto the target tissue or cells using a particle gun (also called as gene gun/shot gun/micro projectile gun).The micro projectile bombardment method was initially named as ballistics by its inventor Sanford (1988). Two types of plant tissue are commonly used for particle bombardment- Primary explants and the proliferating embryonic tissues.

Transformation - This method is used for introducing foreign DNA into bacterial cells e.g. E. Coli. The transformation frequency (the fraction of cell population that can be transferred) is very good in this method. E.g. the uptake of plasmid DNA by E. coli is carried out in ice cold CaCl2 (0-50C) followed by heat shock treatment at 37-450C for about 90 sec. The transformation efficiency refers to the number of transform ants per microgram of added DNA. The CaCl2 breaks the cell wall at certain regions and binds the DNA to the cell surface.

Conjuction - It is a natural microbial recombination process and is used as a method for gene transfer. In conjuction, two live bacteria come together and the single stranded DNA is transferred via cytoplasmic bridges from the donor bacteria to the recipient bacteria.

Liposome mediated gene transfer or Lipofection – Liposomes are circular lipid molecules with an aqueous interior that can carry nucleic acids. Liposomes encapsulate the DNA fragments and then adher to the cell membranes and fuse with them to transfer DNA fragments. Thus, the DNA enters the cell and then to the nucleus. Lipofection is a very efficient technique used to transfer genes in bacterial, animal and plant cells.

SELECTION OF TRANSFORMED CELLS FROM UNTRANSFORMED CELLS

The selection of transformed plant cells from untransformed cells is an important step in the plant genetic engineering. For this, a marker gene (e.g. for antibiotic resistance) is introduced into the plant along with the transgene followed by the selection of an appropriate selection medium (containing the antibiotic). The segregation and stability of the transgene integration and expression in the subsequent generations can be studied by genetic and molecular analyses (Northern, Southern, Western blot, PCR).

Integration of plant tissue culture into plant transformation protocols:

Various methods of plant regeneration are available to the plant biotechnologist. Some plant species may be amenable to regeneration by a variety of methods, but some may only be regenerated by one method. In Chapter 3, the various methods that can be used to transform plants will be considered, but it is worthwhile briefly considering the interaction of plant regeneration and transformation methodologies here (Figure). Not all plant tissue is suited to every plant transformation method, and not all plant species can be regenerated by every method. There is therefore a need to find both a suitable plant tissue culture/regeneration regime and a compatible plant transformation methodology.

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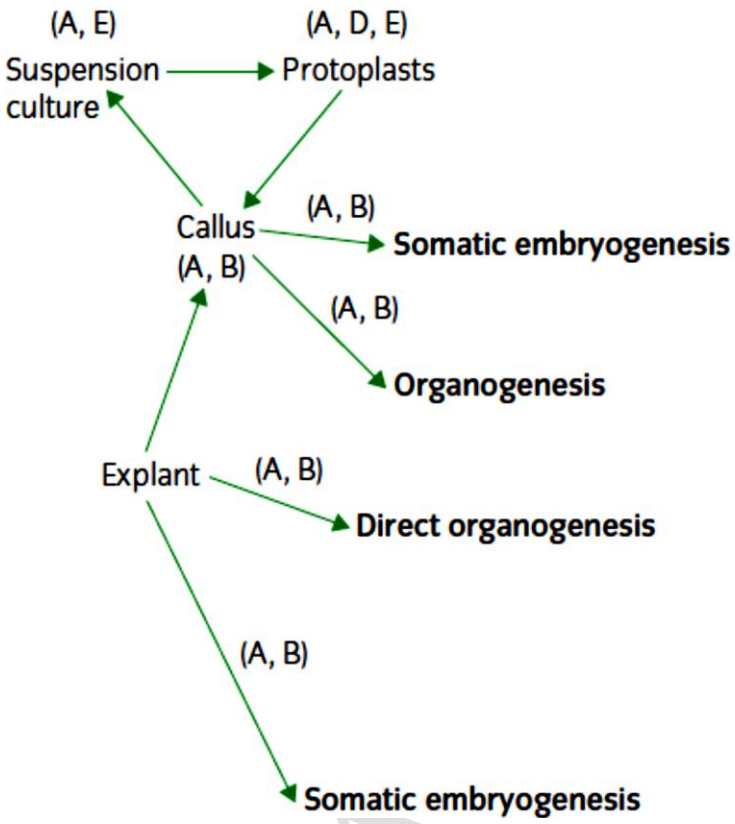
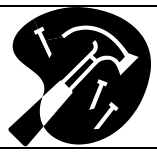


Figure: A simplified scheme for the integration of plant tissue culture into plant transformation protocols. An explant can be a variety of tissues, depending on the particular plant species being cultured. The explant can be used to initiate a variety of culture types, depending on the explant used. Regeneration by either organogenesis or somatic embryogenesis results in the production of whole plants. Different culture types and regeneration methods are amenable to different transformation protocols. The transformation protocols highlighted in this figure are (A) Agrobacterium-mediated, (B) biolistic transformation, (D) direct DNA uptake, and (E) electroporation. Different combinations of culture type and transformation protocol are used depending on the plant species and cultivar being used. In some species, a variety of culture types and regeneration methods can be used, which enables a wide variety of transformation protocols to be utilized. In other species, there is effectively no choice over culture type and/or regeneration method, which can limit the transformation protocols that are applicable.

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GATE Questions

GATE BT 2010

Q.1 Determine the correctness or otherwise of the following **Assertion (a)** and the **Reason (r).**

Assertion: Somatic embryogenesis in plants is a two-step process comprising of embryo initiation followed by embryo production.

Reason: Embryo initiation is independent of the presence of 2,4-dichlorophenoxyacetic acid whereas embryo production requires a high concentration of 2,4-dichlorophenoxyacetic acid.

(A) both (a) and (r) are true and (r) is the correct reason for (a)
(B) both (a) and (r) are true and (r) is not the correct reason for (a)
(C) (a) is true but (r) is false
(D) (a) is false but (r) is true

Q.2 Match the promoters listed in Group I with the tissues listed in Group II.

Group I	Group II
P. α-Amylase	1. Endosperm
Q. Glutenin	2. Tuber
R. Phaseotlin	3. Aleurone
S. Palatin	4. Cotyledon

(A) P-3,Q-1,R-4,S-2 (B) P-3,Q-4,R-1,S-2
(C) P-4,Q-2,R-1,S-3 (D) P-1,Q-3,R-2,S-4

GATE BT 2011

Q.3 Hydrated synthetic seeds which are produced by ion exchange reaction involve mixing the somatic embryos in a solution of

(A) sodium alginate and dropping it in a solution of calcium nitrate
(B) calcium alginate and dropping it in a solution of sodium nitrate
(C) calcium alginate and dropping it in a solution of ammonium nitrate
(D) mannitol and dropping it in a solution of sodium nitrate

Q.4 Shoot organogenesis by tissue culture results into

(A) a bipolar structure that has no vascular connection with the explant
(B) a monopolar structure that has a strong connection with the pre-existing vascular tissue of the explant

Career Avenues

(C) a monopolar structure that has no vascular connection with the explants
(D) a bipolar structure that has a strong connection with the pre-existing vascular tissue of the explants

Q.5 'Hairy roots'induced *in vitro* by the infection of *Agrobacterium rhizogenes*, are characterized by
P.a high degree of lateral branching
Q. genetic instability of culture
R. an absence of geotropism
S. poor biomass production

(A) P and R only (B) P and Q only
(C) Q and R only (D) R and S only

GATE BT 2012

Q.6 Determine the correctness or otherwise of the following **Assertion (a)** and **Reason (r).**

Assertion: The production of secondary metabolites in plant cell cultures is enhanced by the addition of elicitors.
Reason: Elicitors induce the expression of enzymes responsible for the biosynthesis of secondary metabolites.

(A) Both (a) and (r) are true but (r) is not the correct reason for (a)
(B) Both (a) and (r) are true but (r) is the correct reason for (a)
(C) (a) is true but (r) is false
(D) (a) is false but (r) is true

Q.7 Determine the correctness or otherwise of the following **Assertion (a)** and **Reason (r).**

Assertion: In direct somatic embryogenesis, embryos are developed without going through callus formation.
Reason: This is possible due to the presence of pre-embryonically determined cells.

(A) Both (a) and (r) are true but (r) is not the correct reason for (a)
(B) (a) is false but (r) is true
(C) (a) is true but (r) is false
(D) Both (a) and (r) are true but (r) is the correct reason for (a)

Answers:						
1 – A	2 – C	3 – A	4 – B	5 – A	6 – B	7 - A

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